The Limits of Electoral Gender Quotas in Rural Local Bodies

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Abstract

A central promise of electoral gender quotas is that they will help women win office after the quotas end. By demonstrating women's competence, reducing prejudice, and expanding the pool of experienced women politicians, quotas may create lasting electoral opportunities. We test this promise using quota assignment across four election cycles spanning sixteen years in more than 34,000 rural local bodies in India. We distinguish effects in the next open election, their duration across later open elections, and the effects of additional reserved terms. The average next-election gain is 1.1 percentage points (95% interval [0.4, 1.7]), against a non-quota win rate of 12.9%. Across later open elections, we find no consistent pattern of growth or decay. Holding the most recent reserved term fixed, a second term adds an estimated −0.3 to 3.4 points. Three reserved terms rather than one yield estimated gains of 4.0–5.3 points, but these gains shrink to near zero and become imprecise in districts where tests do not detect dependence between earlier quota assignments. Male relatives' mediation of access to women officeholders may help explain these limited electoral gains. In a phone audit of roughly 1,000 representatives, we reached the elected officeholder in 10% of quota seats with an answered call, compared with 77% of non-quota seats. Women elected under quotas are also younger, less educated, and less wealthy than non-quota winners. These patterns suggest that holding office may offer women limited opportunities to build independent political careers.

Keywords: Women in Politics, Gender Quotas, India, Local Government

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1 Introduction

Who holds office affects whose interests government serves. In India, reserving local offices for women has shifted public spending toward women's priorities (Chattopadhyay and Duflo, 2004; cf. Ban and Rao, 2008; Gajwani and Zhang, 2015; Green et al., 2026). Women's political representation has also increased the reporting of crimes against women (Iyer et al., 2012), while exposure to women leaders has raised girls' aspirations and educational attainment (Beaman et al., 2012). These instrumental benefits complement the democratic case for women having an equal voice in government. Quotas provide one route to greater representation, with the promise that their effects will extend beyond the reserved term. By demonstrating women's competence, reducing prejudice, and giving women political experience and networks, quotas may increase both voter support for women and the pool of viable women candidates. A central question is whether these changes help women win office after reservation ends.

Evidence on electoral success after quotas end is mixed. Temporary candidate-list quotas in Italy produced modest increases in women's representation after repeal (De Paola et al., 2010), while estimates for village leadership in West Bengal are small or imprecise (Beaman et al., 2009). A larger estimate comes from Mumbai, where women were 17.9 percentage points more likely to win an unreserved seat following reservation, in a study of 118 wards (Bhavnani, 2009). These findings leave unresolved how broadly such gains extend, whether they endure beyond the next election, and whether additional reserved terms strengthen them.

We examine these questions using sixteen years of electoral data from 34,587 Gram Panchayats across four election cycles in Rajasthan and Uttar Pradesh (with additional rural and urban evidence in Appendix G). A constitutional mandate reserves at least one third of local government leadership positions for women (Constitution of India, Article 243D). We exploit quota assignment to distinguish three questions: whether reservation helps women win the next open election, how that effect changes across subsequent open elections, and what additional reserved terms contribute.

We find a small average gain in women's electoral success in the next open election. Giving equal weight to each state and election transition yields 1.1 percentage points (95% interval [0.4, 1.7]), against an equally weighted non-quota win rate of 12.9%. A meta-analysis combining these results with evidence from Mumbai, Delhi, Chandigarh, Bengaluru, and West Bengal yields an estimated average gain of 1.5 percentage points (95% interval [-1.5, 4.5]). We find no clear evidence that effects are lower in rural than in urban settings (Appendix H).

We also examine whether electoral gains persist beyond the first open election. Following the same places through successive open elections, estimates remain small in UP (-0.2 to 2.3 percentage points) and vary widely in Rajasthan (-12.6 to 9.6 points). We find no consistent pattern of growth or decay after reservation ends (Appendix E).

We also ask whether additional reserved terms produce larger electoral gains. Holding the most recent reserved term fixed, a second term adds an estimated -0.3 to 3.4 percentage points, with all 95% confidence intervals including zero. Three terms rather than one yield larger estimates: 5.3 points in Rajasthan and

4.0 in UP. But when we limit the analysis to districts where tests do not detect dependence between earlier quota assignments, these estimates are near zero and imprecise: −2.4 points in Rajasthan (95% interval [-12.3, 7.6]) and 1.4 in UP ([-5.9, 8.6]). The results remain consistent with small gains from additional exposure; the evidence for larger gains after three terms depends on which districts enter the analysis.

Male relatives' mediation of access to women officeholders may help explain why holding office does not translate more fully into subsequent electoral success. In a phone audit covering roughly 1,000 representatives in Rajasthan, we reached the elected officeholder in 10% of quota seats with an answered call, compared with 77% of non-quota seats. Women elected under quotas are also younger, less educated, and less wealthy than non-quota winners. They are less likely to reappear as candidates in the next election when the seat is open to all genders and castes (Appendix K). These patterns are consistent with quotas drawing women with weaker prior political ties into office, or with officeholding offering them limited opportunities to exercise authority, gain experience, and build independent political careers (Valdini, 2019; Brulé et al., 2025; Meguid et al., 2025).

2 Theoretical Framework

For quotas to help women win after reservation ends, holding office must change voters' willingness to elect women or women's ability and willingness to contest. One route is through learning about competence. Voters uncertain about women's ability to govern may use gender as a proxy. Seeing a woman manage public funds, deliver services, and respond to constituents can provide evidence against unfavorable beliefs. Exposure to women leaders has reduced bias in assessments of women's effectiveness, although changes in beliefs need not translate into equally large electoral gains (Beaman et al., 2009). This mechanism depends on what voters observe and whom they credit for the results.

Exposure may also reduce prejudice against women in authority. Familiarity with women leaders could make female leadership more acceptable even without changing beliefs about competence. This differs from learning that women govern effectively: voters can recognize competence while continuing to prefer men. In West Bengal, exposure reduced bias in assessments of effectiveness without eliminating men's preference for male leaders (Beaman et al., 2009). The distinction matters because quotas may change one barrier to electoral support while leaving another intact.

Quotas can also expand the pool of women able and willing to compete. Office can provide political experience, constituent relationships, recognition, and access to networks. These gains may help the elected woman contest again, while her example encourages other women to enter politics. Exposure to women leaders has raised girls' aspirations and educational attainment (Beaman et al., 2012), and women politicians have recruited more women into grassroots party activism (Goyal, 2024). These findings identify routes through which representation could broaden participation; whether they lead to more women contesting and winning later elections remains an empirical question. Returning officeholders and new entrants are distinct sources of any electoral gain.

Holding office need not give women the authority or visibility that these mechanisms require. Male relatives may conduct public business, negotiate with officials, and receive credit for the council's work, a practice commonly described as *sarpanch pati* when the husband assumes this role (Thakur, 2015; Brulé et al., 2025). Such involvement can limit what voters learn about women's competence, what other women see as possible, and the experience officeholders acquire. It can also constrain women's influence over spending and other policy choices. But assistance from a spouse need not eliminate women's agency: women may still influence decisions or make it easier for other women to express their preferences (Beaman et al., 2011). The implications depend on how authority and public credit are shared.

Women may also enter office with fewer of the resources needed to govern independently. Winners of reserved seats can have less schooling and political experience than non-quota winners, even when they are positively selected relative to other women (Ban and Rao, 2008). In Tamil Nadu, female presidents knew less about panchayat rules and had less contact with higher-level officials despite relatively similar schooling; their councils also built fewer schools and roads, which require coordination with those officials (Gajwani and Zhang, 2015). Differences in qualifications vary across offices and settings (Sood, nd), and age or schooling alone does not establish effectiveness. Experience can improve performance (Ban and Rao, 2008; Afridi et al., 2013), but acquiring that experience requires opportunities to participate in governing.

These mechanisms have distinct implications for the timing of electoral gains. Learning, reduced prejudice, and a larger candidate pool could help women in the first election after a quota ends. Gains could endure in later elections if beliefs, networks, and political experience persist. Additional reserved terms could strengthen these changes, although there is no reason to expect each term to contribute equally. Repeated reservation of an office can also elect different women, so it need not provide any one officeholder with additional experience. We therefore distinguish the next-election effect, its duration across later open elections, and the contribution of additional reserved terms.

Limited authority and preparation could weaken each step from electing women to helping women win later. If male relatives perform the public role and receive credit, voters see less evidence of women's leadership, potential candidates see fewer women exercising power, and officeholders have fewer opportunities to build independent political careers. Constraints on administrative knowledge and access to officials may further limit what women accomplish in office. Our phone audit and comparisons of officeholder characteristics help interpret the electoral results in light of these possibilities. The audit records access to the elected representative, while the administrative records describe who enters office. Following winners into the next election provides a separate indication of whether holding office is followed by continued electoral participation (Appendix K).

The strength of these channels may vary across settings. Greater access to education, public institutions, and political networks could make it easier for women to govern independently and build support. Evidence from South India links women's relative performance to political experience and the institutional environment (Ban and Rao, 2008). Yet settings where women already have more political opportunities may leave less room for quotas to change electoral outcomes. Rural–urban differences therefore provide

evidence about variation in quota effects, without identifying which feature of the setting produces that variation. We examine these differences alongside the evidence on persistence (Appendix H).

3 Institutional Background

We study the elected heads of Gram Panchayats (GPs) in Rajasthan and Uttar Pradesh. A GP comprises one or more villages and forms the lowest tier of rural local government, below the block and district councils (Figure 1). Its head, called the *sarpanch* in Rajasthan and the *pradhan* in UP, is directly elected by the GP's voters. The head leads a council of elected ward members with responsibilities for local development, public services, and the administration of government programs (Government of Rajasthan, 1994a; Government of Uttar Pradesh, 1947).

India's 73rd Constitutional Amendment established a minimum reservation of one third of panchayat head positions for women (Government of India, 2025, Article 243D(4)). Rajasthan raised this minimum to 50% in 2008; UP retained the one-third minimum through the 2021 election (Ministry of Panchayati Raj, 2021). In both states, women's reservations operate alongside reservations for Scheduled Castes (SC), Scheduled Tribes (ST), and Other Backward Classes (OBC). A position can therefore be reserved for women of a particular caste category. Only women can contest women-reserved positions, but women can also contest positions without a gender reservation, subject to any caste restriction (Government of Rajasthan, 1994a; Government of Uttar Pradesh, 1947). We call positions reserved for women quota seats and those without a gender reservation non-quota or open seats; either can carry a caste reservation.

State rules determine which GP-head positions are reserved at each election. Rajasthan's election rules provide for the allocation of women's reservations by drawing lots (Government of Rajasthan, 1994b, Rule 9(3)). UP's procedures have changed over time. The published account of the 2005 and 2010 elections describes selection from caste-specific serial lists, with changes in caste reservation allowing consecutive reservations for women (Bose and Das, 2018, p. 206). In 2015, following delimitation, UP assigned reservations afresh using population-based rankings within blocks and caste categories, without carrying forward previous reservation assignments (Chaturvedi et al., 2025, p. 760).

Panchayats normally serve five-year terms (Government of India, 2025, Article 243E). Across elections, a GP-head position may remain reserved for women or become open to women and men. Our panels contain both consecutive reserved terms and reserved terms followed by one or more open elections. These histories allow us to examine women's electoral success after reservation ends and after different lengths of exposure. When a position becomes open, the woman who held it under reservation can compete alongside other eligible candidates; any subsequent increase in women's representation can therefore come from returning officeholders or new entrants.

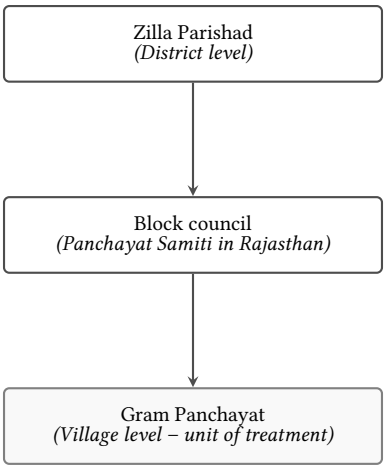


Figure 1: Local Governance Structure in Rajasthan and Uttar Pradesh

Notes: Councils operate at the district, block, and GP levels. A Gram Panchayat comprises one or more villages. The study concerns reservation of the GP-head office.

4 Research Design and Data

4.1 Treatment, outcomes, and comparisons

We estimate the effect of reserving GP-head positions for women on women's subsequent electoral participation and success. The comparison group consists of positions without a women's reservation, including those won by women. The treatment is therefore quota assignment. Its effects can reflect both changes in who enters office and changes in the electoral opportunities and incentives created by reservation. We retain women elected without quotas in the comparison group and estimate the intent-to-treat effect of reservation (Angrist et al., 1996).

Our primary outcome is whether a woman wins an election open to both women and men. The next-election comparison asks whether this outcome differs between GPs reserved for women in the preceding election and GPs without that reservation. Both groups must be open in the outcome election, but either may retain a caste reservation. We also examine women's candidacy in Rajasthan's 2020 election, measuring whether any woman contests, the number and proportion of women candidates, and the votes received by women among the winner and runner-up as a share of all valid votes. These outcomes distinguish entry into electoral competition from winning office.

We separate duration from additional exposure to quotas. To study duration, we hold reservation histories and the GP sample fixed and follow women's electoral success across subsequent open elections. The first comparison follows GPs reserved in 2005 and open thereafter against GPs open in all four observed elections. Two further comparisons follow GPs reserved in 2010, with or without reservation in 2005, through two open elections. To study additional exposure, we compare histories at the same final election: for example, two or three prior reserved terms versus reservation only in the immediately preceding term. These comparisons hold the timing of the last reservation fixed while varying earlier reservation. We also

compare three prior reserved terms with none during the observed period. Reservation before 2005 is unobserved, so the latter group need not have been free of quotas before our panel begins.

4.2 Data and analysis samples

We assemble election records from the Rajasthan and Uttar Pradesh State Election Commissions[1] for four cycles each: 2005, 2010, 2015, and 2020 in Rajasthan, and 2005, 2010, 2015, and 2021 in UP. The records identify the reservation status of the GP-head position and the winner. Winner sex comes from the election records where available; for Rajasthan 2015, it was manually coded from names. Rajasthan's 2020 candidate records supply the candidacy outcomes and officeholder characteristics. We link records within the same election phase and GP, retaining unresolved sex and candidate counts as missing where needed for an outcome. Appendix A documents coverage and construction. Separate analyses use additional settings (Appendix G) and link winners to subsequent candidates (Appendix K); the phone audit is described in Section 6.1.

We construct adjacent-election panels by matching GPs within harmonized districts and blocks, resolving spelling differences and administrative renaming. Ambiguous links remain unmatched. Four-election panels join the adjacent links through the same intermediate election records. Coverage varies across years, and matching does not establish that GP boundaries remained unchanged. Table 1 distinguishes the linked panels from the next-election analysis samples, which additionally require an open outcome election, known prior reservation, an observed outcome, and a block identifier. Duration comparisons require the same histories and observed winner sex at every follow-up; their samples are consequently smaller than the adjacent-election samples. The cumulative models require the full observed reservation history and an open final election.

Women won 4.5–16.5% of non-quota offices in the treatment election among control GPs with known winner sex (Table 1). Most control GPs therefore elected men in the treatment election, but the comparison remains between quota and non-quota assignment. These contemporaneous shares describe the control group's composition; women's win rates in the later outcome election provide the baseline for interpreting persistence.

For balance tests and covariate-adjusted models, we attach 2001 Census population and village characteristics through the Local Government Directory and SHRUG 2.0 (Asher et al., 2021; Office of the Registrar General and Census Commissioner, 2011). Village records are aggregated to GPs using the geographic crosswalks described in Appendix A. The adjusted specifications include log population, SC share, literacy, female population share, and an index of education, medical, electricity, and banking availability. These models use the subset with the required covariates; the main specifications do not require a Census match.

[1] Election records: https://sec.rajasthan.gov.in/ and https://sec.up.nic.in/.

Table 1: Next-election samples and women elected without quotas

	GP counts				Non-quota offices in t	
Elections	Linked	Analyzed	Quota	Non-quota	Sex observed	Women (%)
Rajasthan						
2005–2010	7,667	4,061	1,341	2,720	2,720	4.5
2010–2015	7,447	3,867	2,584	1,283	1,283	10.8
2015–2020	7,882	4,014	2,504	1,510	1,510	6.9
Uttar Pradesh						
2005–2010	41,890	28,002	13,343	14,659	14,537	11.7
2010–2015	39,529	25,963	9,474	16,489	16,482	14.2
2015–2021	46,514	30,975	13,310	17,665	17,665	16.5

Notes: Elections run from treatment year t to outcome year $t + 1$. Linked counts refer to the adjacent-election panels. Analyzed counts require an open seat in $t + 1$ and observed treatment, outcome, and block. Quota and non-quota counts partition this analysis sample by assignment in t. The final two columns describe non-quota offices in t: the number with known winner sex and the percentage won by women among them. Missing winner sex in t does not exclude a GP from a model of winning in $t + 1$. These shares are not the later-election control means. Source: linked state election records.

4.3 Identification

A causal interpretation requires the compared reservation assignments to be independent of potential electoral outcomes within the relevant allocation groups and analysis sample. Rajasthan's rules specify lotteries for women's reservations, providing the basis for this comparison (Government of Rajasthan, 1994b, Rule 9(3)). The eligible offices can depend on caste and reservation history, so a lottery need not produce independent assignments across elections. UP requires separate consideration: the earlier serial-list account and the population rankings used in 2015 describe different procedures (Bose and Das, 2018; Chaturvedi et al., 2025). For UP, the within-block comparisons additionally require that the allocation criteria do not leave differences in potential outcomes between the compared groups. Block fixed effects alone do not establish that condition.

Restricting attention to subsequent open elections creates a further requirement. Past reservation can influence whether a GP becomes open, and additional eligibility rules can affect which histories we observe. Random initial assignment can remain informative under rotation when later eligibility depends only on prior assignments and the controlled allocation groups, with both comparison histories possible. If eligibility also depends on other predictors of women's electoral success, the selected groups can differ for reasons beyond quota assignment. Linkage and missing outcomes can create similar selection. The longer-history comparisons require these conditions over more elections. The policy effect includes behavioral responses to expected future eligibility; public knowledge of the rules does not itself establish comparability.

The assignment diagnostics show little association between successive reservations in Rajasthan's earliest transition and substantial negative association in later transitions; UP also shows dependence across elections (Appendix Tables B1–B4). These patterns describe the reservation histories available for comparison. Dependence does not by itself rule out a constrained lottery, and a failure to reject independence does

not establish random assignment. We use district restrictions based on these tests as sensitivity analyses, reporting their consequences alongside the corresponding electoral estimates.

Balance tests in the open-election samples jointly assess the five Census covariates used in the adjusted models. Conditional on block fixed effects, these tests do not reject balance in any Rajasthan transition but reject it in all three UP transitions (Table B8). They cover the subset with complete Census data. The broader Census tests, which include GPs reserved at follow-up and additional characteristics, reject balance for Rajasthan's first transition ($p = 0.019$) and all three UP transitions (Tables B6 and B7). Prior-outcome tests ask whether future reservation predicts women's earlier electoral success among seats open in that earlier election; some estimates differ from zero (Table B9). Together with the assignment rules, these findings motivate the sensitivity analyses below and warrant greater caution in interpreting the UP comparisons causally.

4.4 Estimation and inference

For each adjacent-election pair, we estimate the linear probability model

$$\text{FemaleWinner}_{g,t+1} = \beta\, \text{Quota}_{g,t} + \lambda_d + \varepsilon_{g,t+1}, \tag{1}$$

among GPs open in $t+1$. Here g indexes GPs and d indexes the outcome-year district–samiti in Rajasthan or district–block in UP. Our main estimates include fixed effects for these groups; we also report estimates without fixed effects and with the pre-treatment covariates described above. To distinguish adjustment from sample selection, we compare adjusted and unadjusted models on the same Census-complete observations. We also use treatment-year blocks and then add fixed effects for caste-reservation categories within those blocks. Standard errors cluster by block, using the corresponding year's identifiers. Under the identifying conditions above, β captures the effect of prior quota assignment on women's probability of winning the following open election. Appendix Table C2 reports wild cluster bootstrap inference. Throughout the paper, rates are reported as percentages and effects on those rates, including standard errors and confidence intervals, as percentage points. Counts and other non-rate outcomes retain their stated units.

We summarize the next-election effects by giving equal weight to each state and to each of its three election transitions. We estimate their joint covariance by clustering on geographic blocks harmonized across elections, retaining each comparison's original sample and fixed effects. Appendix C.2 details the calculation and alternatives for weighting and inference.

For duration comparison a and follow-up k, we estimate

$$\text{FemaleWinner}_{g,t(a,k)} = \theta_{a,k}\, \text{Exposed}_{g,a} + \lambda_{d,a,k} + \varepsilon_{g,a,k}. \tag{2}$$

Here $\text{Exposed}_{g,a}$ identifies the specified reservation history rather than the comparison history, and $t(a,k)$ is the kth consecutive open election after reservation ends. These models use fixed 2005 block groups for both fixed effects and clustering, keeping the GPs and within-block weights constant across follow-ups.

We test changes, $\theta_{a,k} - \theta_{a,1}$, using paired outcomes from the same GPs. Appendix E gives the samples, weights, bootstrap inference, and adjustment for multiple temporal comparisons.

The additional-term models regress winning in the final open election on reservation in each of the three prior elections and all interactions, with outcome-year block fixed effects and clustering. Linear combinations of coefficients compare the specified histories, with uncertainty calculated from their joint covariance matrix. We report the numbers of GPs in each history and blocks containing both histories alongside the contrasts. The candidacy analyses use the same full-history specification for Rajasthan in 2020. Sensitivity analyses and precision assessments accompany each set of results.

5 Results

5.1 Women's Electoral Success after Reservation Ends

Prior reservation produces a small average gain in women's probability of winning the next open election. Giving equal weight to each state and each of its three election transitions yields an estimated increase of 1.1 percentage points (95% interval [0.4, 1.7]), against an equally weighted non-quota win rate of 12.9%. Figure 2 shows this average and the six component estimates. The interval accounts for dependence across elections within the same geographic block. This average summarizes the two states and observed transitions; it does not assume a common effect across them.

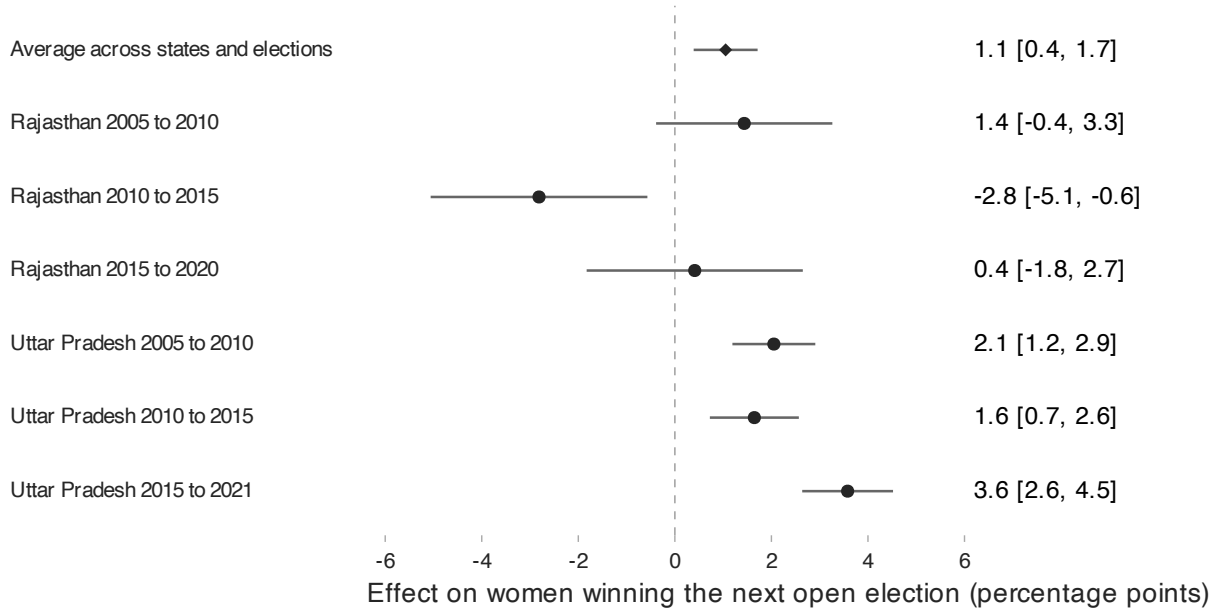


Figure 2: Women's Electoral Success in the Next Open Election

Notes: Estimates compare prior quota and non-quota assignment among GPs not reserved for women at the following election. Circles show the six estimates with outcome-year block fixed effects; the diamond gives equal weight to each state and transition. Lines and brackets show 95% cluster-robust t intervals. The pooled interval uses the joint covariance across elections, clustering on harmonized geographic blocks. Controls include all non-quota GPs, including those previously won by women. Appendix Table C1 reports the regression specifications.

Sensitivity. The estimated average remains modest under alternative weights and sample composition. Weighting the six estimates by their analysis sample sizes gives 2.1 percentage points. Omitting one state–election comparison at a time, while retaining equal weights for the two states, gives averages from 0.6 to 1.7 points. District clustering and a joint wild multiplier bootstrap give similar intervals to the primary calculation (Table C3). The positive average comes from UP: its state average is 2.4 points, compared with -0.3 in Rajasthan. Pooling therefore summarizes variation rather than establishing gains in both states.

Adjusting for the five Census characteristics changes individual estimates by at most 0.2 points on the same observations. Independent UP election data also yield similar estimates. Restricting the sample to districts where tests do not detect dependence between successive quota assignments changes some comparisons and leaves no eligible districts for UP's final transition. Appendix C reports these checks. They do not resolve the allocation and balance concerns discussed in Section 4.3.

Published next-election estimates from West Bengal are also small or imprecise (Beaman et al., 2009). Appendix G presents these estimates alongside the urban evidence and examines sensitivity to which settings enter the synthesis.

5.2 Additional Reserved Terms

Longer reservation could produce larger gains by giving voters more experience of women leaders and women more time to build political careers. Adding a second reserved term yields small, imprecise estimated increments: -0.3 to 3.4 percentage points across the two states and timings, with every 95% interval including zero (Table 2). These comparisons add reservation in either 2005 or 2010 to reservation in 2015, holding the most recent reserved term and the final open election fixed.

Three reserved terms rather than one yield larger full-sample estimates: 5.3 points in Rajasthan and 4.0 in UP. These gains are sensitive to which districts support the comparison. When we limit inference to districts where tests do not detect dependence between quota assignments in the first two transitions, the estimates fall to -2.4 points in Rajasthan (95% interval [-12.3, 7.6]) and 1.4 in UP ([-5.9, 8.6]). The intervals include zero and substantial positive gains. Together, these results do not establish that extending reservation reliably produces larger electoral gains. Appendix D details the restrictions and the separate comparison of three reserved terms with no observed reservation.

Supplementary evidence from Bihar concerns withdrawal after a two-election reservation regime. The descriptive difference in women's winning rates is 3.0 percentage points (95% interval [-1.1, 7.1]; Appendix G). This comparison does not isolate the incremental effect of a second reserved term.

Table 2: Electoral Gains from Additional Reserved Terms

	Rajasthan (2020)		Uttar Pradesh (2021)	
Added reservation	Full	Restricted	Full	Restricted
One extra term, separated	3.4	-2.0	-0.3	-0.9
(2005 added)	[-0.7, 7.4]	[-13.5, 9.2]	[-2.4, 1.8]	[-6.5, 4.6]
One extra term, consecutive	1.9	-1.2	1.2	-1.5
(2010 added)	[-3.2, 6.8]	[-10.6, 8.0]	[-1.5, 3.8]	[-7.1, 4.3]
Two extra terms	5.3	-2.4	4.0	1.4
(2005 and 2010 added)	[-1.0, 11.7]	[-12.3, 7.6]	[0.8, 7.1]	[-5.9, 8.6]

The reference history is reservation only in 2015. Rows add the indicated reserved elections; all outcomes are measured at the final open election, in 2020 for Rajasthan and 2021 for UP. Estimates are in percentage points, with 95% wild cluster bootstrap intervals beneath. Models include all interactions of the three prior assignments and outcome-year block fixed effects, with clustering by block. Restricted samples retain districts where tests do not detect dependence between quota assignments in either 2005–2010 or 2010–2015. Appendix Table D1 reports sample sizes and support.

5.3 Persistence across Later Open Elections

Electoral gains could persist, fade, or emerge only after several open elections. To distinguish these possibilities, we follow GPs reserved in 2005 and open thereafter, comparing them with GPs open in all four observed elections. The same GPs and block weights contribute to each follow-up. Requiring three consecutive open elections and observed outcomes throughout leaves smaller samples than the next-election analysis: 329 GPs in Rajasthan and 6,825 in UP.

The estimates show no consistent growth or decay (Figure 3). UP's estimated differences remain small across all three open elections, while Rajasthan's trajectory is less precisely estimated. From the first to the third open election, the estimated change is -0.3 percentage points in UP (95% interval [-2.8, 2.2]) and 1.8 in Rajasthan ([-8.5, 12.1]). These tests account for repeated observations of the same GPs. They do not establish that the differences remain constant, and Rajasthan's interval permits substantial changes in either direction.

Sensitivity. Relaxing these follow-up requirements leaves UP's estimates small but changes Rajasthan's trajectory. Allowing the sample to vary across follow-ups also changes the GPs being compared, so those estimates cannot establish how a difference evolves for a fixed group. Other reservation histories show larger changes in Rajasthan, without establishing a common pattern across histories or states. Appendix E reports these comparisons, direct tests of changes, and sensitivity to individual blocks.

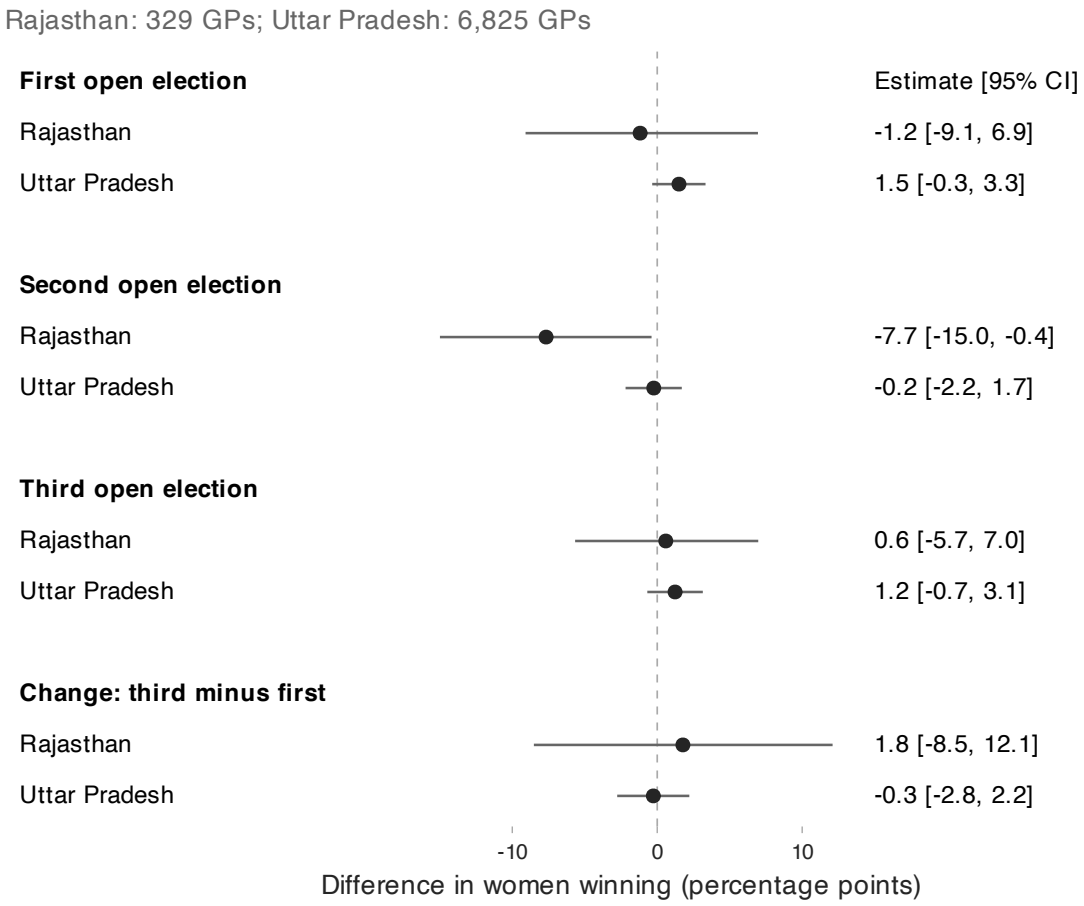


Figure 3: Women's Electoral Success across Three Open Elections

Notes: GPs reserved for women in 2005 and open in all later elections are compared with GPs open in all four observed elections. Each state retains the same GPs across follow-ups, with 2005 block fixed effects and clustering. Points show percentage-point contrasts; horizontal lines show pointwise 95% wild cluster bootstrap intervals. Follow-ups are 2010, 2015, and 2020 in Rajasthan and 2010, 2015, and 2021 in UP. The final group reports third-minus-first contrasts using paired outcomes and the same bootstrap procedure. Caste reservations may remain in either group. Appendix E reports sample support and the full estimates.

5.4 Women's Candidacy

Quotas could encourage more women to run without substantially increasing their chances of winning (O'Connell, 2020; Auerbach and Ziegfeld, 2016). Rajasthan's 2020 candidate records provide little evidence of such an increase. Comparing seats reserved only in 2015 with seats unreserved in all three observed prior elections, the estimated effect on women's share of candidates is 0.7 percentage points (95% interval [-1.9, 3.4]). The estimated effect on whether any woman contests is -0.7 points [-8.2, 6.9], and the estimated change in the number of women candidates is approximately zero [-0.2, 0.2] per GP. All comparisons concern seats open to women and men in 2020 (Appendix Table F1).

The estimated increase in votes received by women among the winner and runner-up is 2.1 percentage points [-0.5, 4.7], measured as a share of all valid votes. This measure excludes votes for women finishing below second place. The estimates therefore provide no clear evidence of increased candidacy or electoral support, although their uncertainty permits meaningful gains.

5.5 Evidence across Rural and Urban Settings

The broader evidence also points to modest average gains after reservation ends. Combining next-election comparisons from Rajasthan, UP, West Bengal, Mumbai, Delhi, Chandigarh, and Bengaluru gives a pooled estimate of 1.5 percentage points (95% interval [-1.5, 4.5]). We first average repeated election comparisons within each setting, then combine the seven setting averages using a random-effects model. Omitting one setting at a time yields estimates between 0.4 and 2.4 points, although their precision varies substantially. Appendix G presents the estimates and sensitivity analyses.

We find no clear rural–urban difference. The fitted persistence contrasts are 1.3 percentage points in Rajasthan, UP, and West Bengal and 2.6 points in Mumbai, Delhi, Chandigarh, and Bengaluru. The rural-minus-urban difference is -1.3 points (95% interval [-11.3, 8.7]; $p = 0.745$). The urban comparisons vary: Bengaluru yields -3.9 points, and Chandigarh's later election yields 0.0 points, both imprecisely estimated. Omitting Bengaluru gives a rural–urban difference of -6.8 points; omitting Mumbai gives 1.3 points. Both intervals include zero. Differences in office type, reservation rules, and election periods also prevent these comparisons from isolating the role of urbanization.

Development within rural settings. Within Rajasthan and UP, we likewise find no clear evidence that quota effects are larger in more developed villages. We examine differences by proximity to towns, infrastructure availability, and female literacy, using measures from the 2001 Census. The estimates do not establish a consistent relationship between development and subsequent electoral gains, but their uncertainty permits meaningful differences. Appendix H.3 reports these comparisons and sensitivity to the distance measure.

6 Officeholding and Political Careers

Quotas could generate lasting electoral gains by giving women opportunities to demonstrate their abilities and acquire political experience. Those opportunities may be limited when male relatives act on their behalf. We examine who handles contact directed to elected representatives, the characteristics of quota and non-quota officeholders, and whether they contest another election.

6.1 Male Proxy Governance

Direct engagement with constituents gives women opportunities to demonstrate their abilities as representatives. In 2024, we conducted a phone audit of Rajasthan's elected village heads using the telephone numbers recorded in their 2020 election filings.[2] We sampled 500 representatives from women-reserved seats and 507 from non-quota seats, including seats won by women. Interviewers asked to speak with the elected representative and made up to three attempts.

Among answered contacts, we reached the elected officeholder in 9.9% of quota seats, compared with 76.7% of non-quota seats (Table 3). These figures include contact through transfers and callbacks. Calls were answered for 382 quota-seat representatives and 404 non-quota representatives, giving response rates of 76% and 80%, respectively.

Male relatives frequently handled calls directed to quota-elected women, and recorded transfers to the officeholder were rare (Appendix Table I1). In one case, the elected representative answered but passed the call to her husband when questions about panchayat governance began. In another, a father-in-law identified himself as the representative. When relatives handle these interactions, voters have fewer opportunities to observe women's leadership, and women have fewer opportunities to develop experience through direct engagement. These patterns complement evidence on male household members as constraints on women's political participation (Cheema et al., 2023; Khan, 2021; Prillaman, 2023; Amar et al., 2024).

The contact gap survives uncertainty about respondent identity. Assigning unresolved cases alternatively to officeholders or other respondents gives contact rates of 9.4–13.4% in quota seats and 74.3–80.9% in non-quota seats (Appendix Table I2). Telephone access and household practices may contribute to this gap. The audit documents who handles calls to the listed number; it does not measure women's participation in meetings or decisions over spending.

6.2 Candidate Characteristics

Opportunities to demonstrate leadership may also depend on the resources women bring to office. Education, financial resources, and experience outside the household could help representatives navigate government and act independently of relatives. Differences in these characteristics may therefore matter for whether reserved terms allow women to establish themselves as political leaders.

[2]The study protocol was reviewed by the IRB under protocol number UP-24-00498.

Table 3: Reaching Elected Representatives by Phone

Outcome	Quota seats	Non-quota seats
Sampled representatives	500	507
Answered: N (% of sampled)	382 (76.4)	404 (79.7)
Officeholder reached: N (% of answered)	38 (9.9)	310 (76.7)

Notes: Phone audit of Rajasthan village heads elected in 2020, conducted in 2024. The unit is a sampled officeholder. Interviewers made up to three attempts; officeholder contact includes transfers and callbacks. Answered wrong numbers count as answered contacts. Non-quota seats include seats won by women. Appendix Table I1 reports respondent identities, relationships, and transfers; Appendix Section I.1 gives the survey protocol.

Table 4: Candidate Characteristics: Quota vs. Open Seats (Rajasthan 2020)

	Winners			Candidates		
Characteristic	Open	Quota	Difference	Open	Quota	Difference
Age	42.8	41.3	1.6	41.5	39.9	1.7
Graduate (%)	23.9	12.7	11.2	22.2	13.3	8.9
Unemployed (%)	9.9	60.9	-50.9	10.7	61.4	-50.7
Log assets	12.9	12.6	0.4	12.3	12.1	0.3

Note:
Descriptive open-minus-quota comparisons, conditional on candidacy or winning. Graduation and unemployment are percentages; their differences are percentage points. Other rows use their stated units. Graduation rates exclude records with missing education. Age means exclude recorded ages outside 21–110 years; other outcomes retain those records. Assets winsorized at the 10th and 90th percentiles of the candidate sample before taking logs. These comparisons do not identify a causal effect on candidate quality.

In Rajasthan's 2020 election, quota-seat winners are younger, less educated, and report fewer assets than non-quota winners, including women (Table 4). Their average age is 41.3 years, compared with 42.8 in non-quota seats; 12.7% are graduates, compared with 23.9%. Mean log assets are 12.6 versus 12.9, and 60.9% report being unemployed, compared with 9.9%. The candidate pool shows differences in the same direction. UP's winner records also show lower average age and education in quota seats across three elections (Appendix Table J2).

These comparisons describe the people elected under the two reservation regimes. They do not establish differences in governing ability, and reported unemployment need not imply an absence of work or relevant experience. The overall gaps also reflect the different sex composition of quota and non-quota winners. Among women winners in Rajasthan, graduation rates and mean log assets are similar across the two groups: 12.7% versus 12.2%, and 12.6 in both. Women winning quota seats remain younger and more likely to report being unemployed (Appendix Table J1). The evidence thus identifies differences in the resources officeholders bring to the role, while leaving open how those resources affect their independence and subsequent political careers.

6.3 Contesting Again

Quotas could produce lasting representation by helping women build political careers that continue after reservation ends. Yet quota-seat winners are less likely than non-quota winners to contest the next election in the same GP. We follow 2015 winners to Rajasthan's 2020 election and UP's 2021 election, restricting both samples to seats subsequently reserved for neither women nor a caste group. This ensures that the new reservation category does not prevent prior winners from running. The comparison includes all prior non-quota winners, including women.

Under strict name matching, 2.8% of quota winners reappear as candidates in Rajasthan, compared with 13.5% of non-quota winners. The corresponding rates in UP are 10.0% and 16.5%. Adjusting for prior caste reservation and block fixed effects gives quota-minus-non-quota differences of -11.5 percentage points in Rajasthan (95% interval [-14.6, -8.3]) and -5.7 points in UP [-6.9, -4.6]. More generous matching raises the rates to 4.4% versus 20.5% in Rajasthan and 27.8% versus 47.1% in UP. The levels are sensitive to how names are matched, but quota winners return less often under every rule (Appendix K).

These comparisons suggest that quota winners less often convert a reserved term into continued electoral participation in the same office. This could reflect quotas bringing less politically established women into office, fewer opportunities to build a career while serving, or both. The comparisons do not distinguish those explanations. They also do not capture candidacy elsewhere or the entry of other women.

7 Conclusion

Quotas secure women's representation while seats are reserved, but the electoral gains after reservation ends are modest. Across Rajasthan and UP, the average next-election gain is 1.1 percentage points, against a non-quota win rate of 12.9%. Additional reserved terms do not reliably produce larger gains, and following the same places through later open elections reveals no consistent growth or decay. The broader evidence also points to modest average gains, without establishing that rural effects are smaller than urban effects.

These findings qualify the expectation that electing women will itself generate a growing pool of experienced candidates and greater voter support. In our phone audit, we reached the officeholder in 10% of quota seats with an answered call, compared with 77% of non-quota seats; male relatives frequently handled calls directed to women. Quota winners also entered office with fewer educational and financial resources and were less likely to contest the next election in the same GP. Together, these patterns suggest that access to office may provide limited opportunities to demonstrate leadership and build an independent political career.

The democratic value of women's representation does not depend on quotas eventually becoming unnecessary. But the expectation that reserved terms will generate substantial electoral gains afterward requires more than women's presence in office: it requires opportunities to exercise authority, gain recognition, and continue in politics.

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Supplementary Information for "The Limits of Electoral Gender Quotas in Rural Local Bodies"

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A Data, Measurement, and Sample Construction

A.1 Election coverage and measurement

Table A1 reports the GPs, blocks, and districts identified in each election file. GP counts use distinct, complete district–block–GP names after cleaning; reservation percentages exclude unknown or conflicting status. These are counts in the collected records, not a complete census of administrative units in every state-year. The main next-election samples additionally require known reservation status in both elections, no women's reservation at follow-up, a measured follow-up outcome, and a block identifier (Table 1). Winner sex in the treatment election is not an eligibility condition for these models. Table 1 therefore reports the number of non-quota offices with known contemporaneous winner sex separately from the full control-group count.

For Rajasthan's 2005 and 2010 elections, winner sex is recorded in the source files. The 2015 variable was manually coded from winner names. For 2020, winner and candidate records are joined within the same GP and general-election phase. Unresolved candidate identities or sex remain missing where required to construct an outcome. UP winner sex is carried through from the source election records. These measures describe recorded or inferred sex, and the Rajasthan 2015 results depend on the name coding.

A.2 Linking election records

Constructing the panels requires matching administrative units whose names vary because of spelling differences, transliteration, digitization errors, and administrative renaming. We standardize district and block names using geographic crosswalks, including historical names and changes in district membership. Rajasthan GP links use the historical district–samiti–GP crosswalks. In UP, matching preserves Hindi vowel marks, replaces punctuation with spaces, and excludes numerically conflicting names. It first establishes mutually unique block correspondences within harmonized districts, then accepts GP links that are uniquely nearest in both directions with Jaro distance below 0.1 in Hindi or English. Ties and competing close identities across scripts remain unmatched.

Each adjacent-election pair is linked separately. Four-election histories join adjacent pairs through the same intermediate source-record identities rather than matching the first and last elections independently. Unknown reservation status is excluded from the analysis panels, and model-specific exclusions for missing outcomes follow linkage. A matched GP name does not establish unchanged territorial boundaries; coverage and linkage losses limit the population represented by the estimates.

A.3 Census covariates

The geographic crosswalks attach election GPs to Local Government Directory codes and then to SHRUG village identifiers (Asher et al., 2021). We aggregate the 2001 Primary Census Abstract and Village Directory to GPs. Population counts are summed across linked villages, and demographic shares are calculated from the corresponding GP totals. Facility availability records whether a linked village has the facility;

an all-zero classification requires complete observed coverage. The infrastructure index used in adjusted models sums availability indicators for education, medical care, electricity, and banking. Entirely missing counts remain missing. A missing geographic or Census match does not remove an election row from the main panel; models requiring those covariates use complete cases.

A.4 Election coverage

Table A1: Summary of Gram Panchayats by Election Year

	2005	2010	2015	2020/2021
Panel A: Rajasthan				
Identified GPs	9,156	9,166	9,862	11,311
Blocks	237	249	300	352
Districts	32	33	33	33
Women reserved (%)	33.5	47.6	48.4	48.6
Panel B: Uttar Pradesh				
Identified GPs	51,404	51,757	58,994	49,772
Blocks	842	843	823	728
Districts	70	72	75	67
Women reserved (%)	46.6	33.1	33.8	33.7

Note:
Identified GPs are distinct complete district–block–GP names in the source files, using cleaned GP names. Missing identities are excluded from GP counts; repeated names within a block count once. Reservation shares use identified GPs with known, non-conflicting reservation. District and block counts use observed names. The last column contains Rajasthan 2020 and UP 2021. Coverage changes reflect both administrative changes and source availability.

A.5 Independent Uttar Pradesh data

The independent UP data assembled by Weaver (2025) for Narasimhan and Weaver (2024) cover 61,338 Gram Panchayats across the 2010, 2015, and 2021 elections. The last wave is coded 2020 in the source. The records include reservation status, winner sex, age, education, assets, turnout, and candidate-pool composition. Electoral comparisons use 2011 Census district–block identifiers for fixed effects and clustering. The absence of 2005 records prevents replication of the three-prior-term comparison. Appendices C, D, and J report the relevant estimates.

B Quota Assignment and Identification

Quota assignments vary across elections, and our panels include both consecutive reservations and transitions out of reservation. We present four types of evidence on identification: treatment rotation regressions, chi-squared tests of independence, balance tests on prior electoral characteristics, and balance tests on pre-treatment census covariates.

B.1 Assignment across elections

Table B1: Treatment Rotation: Quota Assignment Correlation Across Election Cycles (2-Way Panel)

	Rajasthan						Uttar Pradesh					
	05→10		10→15		15→20		05→10		10→15		15→21	
	No FE	FE	No FE	FE	No FE	FE	No FE	FE	No FE	FE	No FE	FE
	[i]	[ii]	[iii]	[iv]	[v]	[vi]	[vii]	[viii]	[ix]	[x]	[xi]	[xii]
Intercept	46.7***		67.3***		62.3***		34.3***		37.7***		42.8***	
	(0.6)		(1.2)		(1.2)		(0.3)		(0.2)		(0.2)	
$Quota_{t-1}$	1.1	1.1	-40.5***	-40.9***	-29.0***	-29.0***	-2.5***	-2.6***	-10.2***	-10.3***	-28.0***	-28.1***
	(1.2)	(1.2)	(2.5)	(2.6)	(2.5)	(2.5)	(0.6)	(0.6)	(0.5)	(0.6)	(0.4)	(0.4)
R^2	0.00	0.01	0.16	0.17	0.08	0.09	0.00	0.01	0.01	0.01	0.08	0.08
Observations	7,667	7,667	7,447	7,447	7,882	7,882	41,890	41,890	39,529	39,529	46,514	46,514
Block fixed effects	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes

***p<0.01; **p<0.05; *p<0.1. Outcome: quota status in election t. Treatment: quota status in election $t-1$. Sample includes all GPs (not restricted to open seats). A negative coefficient indicates rotation away from previously reserved GPs, consistent with the rotation mandate. Standard errors clustered by district-samiti (Rajasthan) or district-block (UP). Coefficients and standard errors are in percentage points; intercepts are percentages.

Table B2: Treatment Rotation: Quota Assignment Correlation Across Election Cycles (4-Way Panel)

	Rajasthan						Uttar Pradesh					
	05→10		10→15		15→20		05→10		10→15		15→21	
	No FE	FE	No FE	FE	No FE	FE	No FE	FE	No FE	FE	No FE	FE
	[i]	[ii]	[iii]	[iv]	[v]	[vi]	[vii]	[viii]	[ix]	[x]	[xi]	[xii]
Intercept	46.1***		68.1***		64.3***		34.1***		37.7***		43.8***	
	(0.7)		(1.4)		(1.5)		(0.4)		(0.3)		(0.2)	
$Quota_{t-1}$	2.0	2.0	-42.2***	-42.8***	-31.7***	-32.0***	-2.1***	-2.2***	-10.5***	-10.6***	-28.1***	-28.2***
	(1.4)	(1.4)	(2.7)	(2.8)	(2.9)	(2.9)	(0.7)	(0.7)	(0.6)	(0.6)	(0.5)	(0.5)
R^2	0.00	0.03	0.18	0.19	0.10	0.12	0.00	0.01	0.01	0.02	0.08	0.09
Observations	5,334	5,334	5,334	5,334	5,334	5,334	29,253	29,253	29,253	29,253	29,253	29,253
Block fixed effects	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes

***p<0.01; **p<0.05; *p<0.1. Outcome: quota status in election t. Treatment: quota status in election $t-1$. Sample includes GPs present in all four election years (4-way panel). A negative coefficient indicates rotation away from previously reserved GPs, consistent with the rotation mandate. Standard errors clustered by district-samiti (Rajasthan) or district-block (UP). Coefficients and standard errors are in percentage points; intercepts are percentages.

Tables B1 and B2 show how reservation in one election predicts reservation in the next. Table B1 uses adjacent-election panels, while Table B2 restricts to GPs present in all four election years. Rajasthan's 2005–2010 association is near zero; its later transitions and all three UP transitions have negative estimates. These coefficients are reported in percentage points: a negative value means that GPs reserved in one election are less likely to be reserved in the next.

Tables B3 and B4 test independence of quota assignment across election pairs. In Rajasthan, the tests do not reject independence for 2005–2010 or the non-adjacent 2005–2015 and 2005–2020 pairs, but reject it for the later adjacent elections. In UP, the tests reject independence for the reported pairs, including the non-adjacent 2005–2021 comparison.

Table B3: Chi-Squared Tests for Treatment Rotation (Rajasthan)

Comparison	Chi-Squared	P-Value
2005-2010	1.80	0.180
2010-2015	946.85	<0.001
2015-2020	536.04	<0.001
2005-2015	0.08	0.771
2005-2020	0.15	0.698
2010-2020	111.30	<0.001

Table B4: Chi-Squared Tests for Treatment Rotation (UP)

Comparison	Chi-Squared	P-Value
2005-2010	14.82	<0.001
2010-2015	318.35	<0.001
2015-2021	2316.86	<0.001
2005-2015	535.59	<0.001
2005-2021	4.93	0.026
2010-2021	69.14	<0.001

B.2 Balance on prior electoral characteristics

Table B5 tests whether quota assignment is balanced on prior electoral characteristics across all election transitions. In Rajasthan, the 2005→2010 transition shows similar rates of prior female winners, but later transitions show imbalance consistent with rotation—reserved GPs were less likely to have had prior female winners. In UP, reserved GPs had slightly fewer prior female winners and differed on caste reservation status across all transitions. The primary models include block fixed effects; Table C5 separately conditions on caste reservation in the treatment election. We do not control for the sex of the winner elected under the quota assignment being studied: that is a consequence of treatment.

Table B5: Electoral Balance Tests for Quota Assignment Across Election Cycles

	Rajasthan									Uttar Pradesh								
	05→10			10→15			15→20			05→10			10→15			15→21		
Variable	Q	O	p	Q	O	p	Q	O	p	Q	O	p	Q	O	p	Q	O	p
Prior female winner	36.7	36.0	0.505	33.4	70.4	< 0.001	38.4	64.5	< 0.001	50.6	53.2	< 0.001	35.9	44.7	< 0.001	28.7	51.7	< 0.001
OBC reservation	15.4	14.7	0.394	17.3	13.8	< 0.001	16.0	12.9	< 0.001	20.9	30.4	< 0.001	26.4	28.8	< 0.001	16.2	32.6	< 0.001
SC/ST reservation	35.2	35.0	0.839	35.8	35.8	0.996	40.4	37.2	0.004	22.4	20.7	< 0.001	15.5	24.6	< 0.001	26.9	18.2	< 0.001
N	7,667			7,447			7,882			41,890			39,529			46,514		

Notes: Balance tests for quota assignment on prior electoral characteristics across all election transitions. Q = Quota (GPs reserved for women in the later year); O = Open (GPs not reserved). Prior female winner = proportion with female sarpanch/pradhan in prior election. Quota and open columns report percentages. p-values from t-tests. Full electoral panel (not SHRUG-filtered).

B.3 Balance on Census characteristics

Tables B6 and B7 report descriptive balance on Census covariates measured before treatment. They include pooled likelihood-ratio tests and joint Wald tests with district–block fixed effects and clustering. The block-adjusted joint tests reject balance for Rajasthan's first transition ($p = 0.019$) and all three UP transitions. These broader samples include GPs reserved at follow-up and use a larger set of Census characteristics than the open-election tests below. The tests account for the analysis's block structure; they are not randomization tests.

Table B6: Rajasthan: Balance Tests for Quota Assignment

	2005→2010			2010→2015			2015→2020		
Variable	Quota	Open	p	Quota	Open	p	Quota	Open	p
Population	4088.1	4118.3	0.475	4069.8	4126.0	0.179	4112.5	4103.0	0.822
Female (%)	48.4	48.3	0.213	48.4	48.4	0.540	48.4	48.4	0.246
SC (%)	18.3	18.1	0.529	18.5	17.7	0.014	18.0	18.0	0.876
ST (%)	14.0	14.4	0.556	14.3	14.7	0.466	14.6	14.6	0.952
Medical facility (%)	84.0	84.9	0.438	85.2	84.0	0.258	84.3	85.3	0.367
Maternity homes/1k	0.0	0.0	0.586	0.0	0.0	0.729	0.0	0.0	0.815
Maternal/child centers/1k	0.0	0.0	0.064	0.0	0.0	0.589	0.0	0.0	0.519
Primary schools/1k	1.1	1.1	0.702	1.1	1.1	0.057	1.1	1.1	0.068
Middle schools/1k	0.4	0.4	0.038	0.4	0.4	0.658	0.4	0.4	0.676
Banking (%)	21.8	21.8	0.979	21.2	22.2	0.414	22.1	21.6	0.682
Power supply (%)	98.6	97.9	0.063	98.3	98.3	0.993	98.2	98.4	0.609
Mud road (%)	85.9	86.7	0.492	86.4	86.5	0.914	85.9	87.0	0.270
N		4980			4605			4514	
Joint p (LR)		0.247			0.159			0.598	
Joint p (block)		0.019			0.132			0.301	

Notes: Balance tests for quota assignment across all Rajasthan panels. "Quota" = GPs reserved for women in base year; "Open" = GPs not reserved. Covariates from the 2001 Census Village Directory (SHRUG). Availability is the percentage of GPs with an observed facility in at least one linked village; absence requires complete observed coverage. Counts per 1k use GP population. Maternity/child centers are maternity and child welfare centers. Water-source code sums are excluded because they do not identify facility counts or availability. p-values from t-tests. Joint p (LR) is a descriptive pooled logistic likelihood ratio test. Joint p (block) is a Wald test with district-samiti/block fixed effects and standard errors clustered at that level. These are balance diagnostics, not randomization inference.

Table B7: Uttar Pradesh: Balance Tests for Quota Assignment

	2005→2010			2010→2015			2015→2021		
Variable	Quota	Open	p	Quota	Open	p	Quota	Open	p
Population	2271.5	2404.9	< 0.001	2330.8	2336.7	0.813	2780.2	2144.9	< 0.001
Female (%)	47.5	47.3	< 0.001	47.2	47.3	0.028	47.3	47.3	0.395
SC (%)	22.5	24.4	< 0.001	24.8	23.0	< 0.001	22.4	23.3	0.008
ST (%)	0.1	0.1	0.313	0.0	0.1	0.018	0.1	0.1	0.549
Medical facility (%)	38.9	41.8	< 0.001	40.3	39.8	0.613	44.3	39.0	< 0.001
Maternity homes/1k	0.0	0.0	0.724	0.0	0.0	0.112	0.0	0.0	0.041
Maternal/child centers/1k	0.1	0.1	0.011	0.1	0.1	0.422	0.1	0.1	0.326
Primary schools/1k	0.7	0.7	0.297	0.7	0.7	0.519	0.7	0.8	< 0.001
Middle schools/1k	0.2	0.2	0.889	0.2	0.2	0.158	0.2	0.2	0.272
Banking (%)	7.8	9.4	< 0.001	8.9	8.5	0.487	12.1	7.1	< 0.001
Power supply (%)	82.5	82.9	0.534	81.3	82.0	0.304	82.1	81.3	0.318
Mud road (%)	68.3	67.3	0.145	65.7	66.7	0.236	64.9	68.4	< 0.001
N		17345			14942			9865	
Joint p (LR)		< 0.001			< 0.001			< 0.001	
Joint p (block)		< 0.001			< 0.001			< 0.001	

Notes: Balance tests for quota assignment across all Uttar Pradesh panels. "Quota" = GPs reserved for women in base year; "Open" = GPs not reserved. Covariates from the 2001 Census Village Directory (SHRUG). Availability is the percentage of GPs with an observed facility in at least one linked village; absence requires complete observed coverage. Counts per 1k use GP population. Maternity/child centers are maternity and child welfare centers. Water-source code sums are excluded because they do not identify facility counts or availability. p-values from t-tests. Joint p (LR) is a descriptive pooled logistic likelihood ratio test. Joint p (block) is a Wald test with district-samiti/block fixed effects and standard errors clustered at that level. These are balance diagnostics, not randomization inference.

Table B8 tests balance in the samples used for the covariate-adjusted electoral models. Each regression predicts treatment-year quota assignment from log population, SC share, literacy, female population share, and the infrastructure index, with outcome-year block fixed effects. The sample requires an open follow-up election, observed treatment and winner sex at follow-up, a block identifier, and all five Census covariates. Joint Wald tests use block-clustered covariance estimates. They reject the joint null in each UP comparison and do not reject it in Rajasthan. These tests address observed characteristics in the Census-complete subset; nonrejection does not establish random assignment.

Table B8: Census Balance in the Open-Election Analysis Samples

Elections	N	Blocks	Joint p-value
Rajasthan			
2005–2010	2,684	230	0.264
2010–2015	2,343	230	0.230
2015–2020	2,279	229	0.695
Uttar Pradesh			
2005–2010	11,607	363	< 0.001
2010–2015	9,904	323	< 0.001
2015–2021	6,517	273	< 0.001

B.4 Temporal falsification

Temporal falsification tests examine whether future treatment predicts past outcomes. A failure to reject does not verify random assignment, and the estimated coefficients in Table B9 are not uniformly zero. The diagnostics use block-clustered inference; we do not permute reservation assignments globally without a documented assignment mechanism.

Table B9: Placebo Tests: Future Treatment Predicting Past Outcomes

	Rajasthan						Uttar Pradesh					
	05→10		10→15		15→20		05→10		10→15		15→21	
	No FE	FE	No FE	FE	No FE	FE	No FE	FE	No FE	FE	No FE	FE
	[i]	[ii]	[iii]	[iv]	[v]	[vi]	[vii]	[viii]	[ix]	[x]	[xi]	[xii]
Intercept	4.5***		10.8***		6.8***		11.7***		14.2***		16.5***	
	(0.4)		(1.2)		(0.8)		(0.4)		(0.4)		(0.4)	
$Quota_t$	-0.3	-0.4	-1.3	-1.0	1.0	1.7*	-0.4	-0.3	-0.3	-0.3	0.1	0.1
	(0.5)	(0.5)	(1.1)	(1.2)	(0.9)	(0.9)	(0.5)	(0.5)	(0.5)	(0.5)	(0.4)	(0.4)
R^2	0.00	0.06	0.00	0.11	0.00	0.15	0.00	0.07	0.00	0.06	0.00	0.05
Observations	5,101	5,101	3,918	3,918	4,061	4,061	22,130	22,130	26,459	26,459	30,893	30,893
Block fixed effects	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes

***p<0.01; **p<0.05; *p<0.1. Placebo test: FUTURE quota status predicting PAST outcomes. Outcome: woman elected in open seat at $t-1$. Treatment: quota status at t. Sample restricted to GPs where seat was open in the outcome election. Non-rejection is a limited temporal diagnostic and does not establish random quota assignment. Standard errors clustered by district-samiti (Rajasthan) or district-block (UP). Coefficients and standard errors are in percentage points; intercepts are percentages.

C Women's Electoral Success in the Next Open Election

The average next-election gain is 1.1 percentage points (95% interval [0.4, 1.7]). This appendix reports the component regressions, the construction of that average, and sensitivity to inference, specification, and sample composition.

C.1 Component estimates and precision

Table C1 reports the six adjacent-election regressions. Table C2 combines sample support, block-clustered and wild-bootstrap inference, and approximate minimum detectable effects. The latter range from 1.2 to 3.2 percentage points in the full samples, using 80% power and two-sided 5% size. These fitted-model precision summaries do not establish that an effect is zero.

Table C1: Women's Electoral Success in the Next Open Election: Regression Specifications

	Rajasthan						Uttar Pradesh					
	05→10		10→15		15→20		05→10		10→15		15→21	
	No FE	FE	No FE	FE	No FE	FE	No FE	FE	No FE	FE	No FE	FE
	[i]	[ii]	[iii]	[iv]	[v]	[vi]	[vii]	[viii]	[ix]	[x]	[xi]	[xii]
Intercept	9.2***		8.8***		12.2***		13.2***		15.3***		18.5***	
	(0.7)		(1.1)		(1.0)		(0.4)		(0.3)		(0.4)	
$Quota_{t-1}$	1.5	1.4	-2.0*	-2.8**	0.5	0.4	2.0***	2.1***	1.7***	1.6***	3.7***	3.6***
	(0.9)	(0.9)	(1.2)	(1.1)	(1.1)	(1.1)	(0.5)	(0.4)	(0.5)	(0.5)	(0.5)	(0.5)
R^2	0.00	0.10	0.00	0.15	0.00	0.12	0.00	0.06	0.00	0.05	0.00	0.05
Observations	4,061	4,061	3,867	3,867	4,014	4,014	28,002	28,002	25,963	25,963	30,975	30,975
Block fixed effects	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes

***p<0.01; **p<0.05; *p<0.1. Outcome: woman elected in open seat. Sample restricted to GPs where seat was not reserved for women in outcome year. Block fixed effects and standard-error clustering use outcome-year district-samiti groups in Rajasthan and district-block groups in UP. Coefficients and standard errors are in percentage points; intercepts are percentages.

Table C2: Next-Election Estimates: Support, Inference, and Precision

State	Elections	Sample	N (L/R)	Mixed blocks	Estimate	Cluster CI	Bootstrap CI	MDE
Rajasthan	2005–2010	full	1341/2720	231	1.4	[-0.4, 3.3]	[-0.4, 3.2]	2.6
Rajasthan	2005–2010	restricted	1341/2720	231	1.4	[-0.4, 3.3]	[-0.4, 3.3]	2.6
Rajasthan	2010–2015	full	2584/1283	233	-2.8	[-5.1, -0.6]	[-5.1, -0.6]	3.2
Rajasthan	2010–2015	restricted	356/421	47	2.1	[-1.1, 5.2]	[-1.1, 5.3]	4.4
Rajasthan	2015–2020	full	2504/1510	275	0.4	[-1.8, 2.7]	[-1.8, 2.6]	3.2
Rajasthan	2015–2020	restricted	657/669	99	0.8	[-2.6, 4.1]	[-2.4, 4.0]	4.7
UP	2005–2010	full	13343/14659	716	2.1	[1.2, 2.9]	[1.2, 2.9]	1.2
UP	2005–2010	restricted	10082/11311	547	2.0	[1.0, 3.0]	[1.0, 3.0]	1.4
UP	2010–2015	full	9474/16489	673	1.6	[0.7, 2.6]	[0.7, 2.6]	1.3
UP	2010–2015	restricted	1803/3438	134	1.3	[-0.7, 3.4]	[-0.7, 3.4]	2.9
UP	2015–2021	full	13310/17665	693	3.6	[2.6, 4.5]	[2.6, 4.5]	1.3

Note:
Estimates, 95% intervals, and approximate minimum detectable effects (MDEs) are in percentage points. All models include block fixed effects and cluster by block. Mixed blocks contain both compared groups; L/R counts refer to the left/right groups in each contrast. L/R counts denote prior quota/non-quota offices. The model sample is their sum. Bootstrap intervals use 9,999 null-imposed wild cluster draws. MDEs use a normal approximation at 80% power and two-sided 5% size.

C.2 Pooling Next-Election Comparisons

We give each state half the weight and each of its three next-election comparisons one-third of the state's weight. The resulting average is $\hat{\beta}_{\text{avg}} = \sum_{j=1}^{6} w_j \hat{\beta}_j$, with $w_j = 1/6$. Each component retains its original analysis sample and outcome-year block fixed effects. The non-quota baseline averages the six comparison-group win rates with the same weights. These weights describe the studied states and transitions, rather than a population-weighted effect for rural India. Within each component, the block-fixed-effects coefficient retains its regression weights.

To estimate dependence across comparisons, we reconcile geographic block names using the linked election records, including district renamings in UP. The observed block mappings are one-to-one across adjacent panels. Let $\tilde{x}_{igj}$ denote treatment demeaned within the block for comparison j, and $\hat{u}_{igj}$ its fitted residual. For geographic block g, define the influence contribution

$$q_{gj} = \frac{\sum_{i \in g} \tilde{x}_{igj} \hat{u}_{igj}}{\sum_i \tilde{x}_{ij}^2}.$$

Blocks absent from a comparison contribute zero. We apply each equation's usual cluster and degrees-of-freedom corrections to its influence contributions, then form the joint covariance $\widehat{V} = Q'Q$. Its diagonal reproduces the six original cluster-robust variances; its off-diagonal entries capture dependence across elections within the same geographic block. The pooled variance is $w'\widehat{V}w$. We use a t reference distribution with degrees of freedom equal to the smallest number of clusters among contributing comparisons minus one. Inference allows arbitrary dependence within blocks across elections and assumes independence across blocks.

Table C3 varies the weights and composition and also allows dependence within districts. The joint wild multiplier bootstrap uses 9,999 Rademacher draws on the corrected block influence contributions, assigning the same multiplier to a block across all elections; its interval uses the quantiles of the centered pooled draws. Omitting a comparison preserves equal state weights and redistributes the omitted comparison's weight across the remaining elections in that state. Omitting a state instead reports the remaining state's average. These checks change the set of effects being averaged and do not estimate a common effect for an unobserved population of states.

Table C3: Sensitivity of the Average Next-Election Effect

Average or inference method	Estimate	95% interval
Equal states and transitions	1.1	[0.4, 1.7]
Analysis-sample weights	2.1	[1.7, 2.6]
District clustering	1.1	[0.3, 1.8]
Omit Rajasthan	2.4	[1.9, 2.9]
Omit Uttar Pradesh	-0.3	[-1.5, 0.9]
Omit Rajasthan 2005–2010	0.6	[-0.2, 1.5]
Omit Rajasthan 2010–2015	1.7	[0.9, 2.4]
Omit Rajasthan 2015–2020	0.9	[0.1, 1.6]
Omit UP 2005–2010	1.1	[0.5, 1.8]
Omit UP 2010–2015	1.2	[0.6, 1.9]
Omit UP 2015–2021	0.8	[0.1, 1.5]
Joint wild multiplier bootstrap	1.1	[0.4, 1.7]

Estimates and 95% intervals are in percentage points. All rows retain the original block-fixed-effects estimates. Unless specified otherwise, intervals use the joint block-cluster covariance across elections. Analysis-sample weights are proportional to the six regression sample sizes, counting a GP again when it appears in a different comparison. The district and bootstrap rows change inference, keeping the primary weights. Each omission averages only the remaining comparisons.

C.3 Specification and sample sensitivity

Table C4 separates Census sample selection from covariate adjustment: it reports the full sample, the complete-Census subset without covariates, and the same subset with the five covariates. Adjustment changes estimates by at most 0.2 percentage points on the same observations.

Table C5 holds the sample fixed while changing the allocation groups. It requires both block identifiers and observed treatment-year caste reservation, but no Census match. Using treatment-year rather than outcome-year blocks leaves the point estimates unchanged. Adding treatment-year block-by-caste-reservation fixed effects changes Rajasthan's 2010→2015 estimate from -2.8 to -3.8 points; other changes are smaller. The treatment-year specifications cluster by treatment-year block. Caste categories describe the office's reservation, not the winner's caste. These adjustments do not resolve imbalance in UP or establish comparability on unobserved characteristics.

The restricted rows in Table C2 retain districts where tests do not detect dependence between successive quota assignments. Rajasthan's 2010→2015 estimate changes from -2.8 to 2.1 percentage points (95%

cluster interval [-1.1, 5.2]; $N = 777$); its 2015→2020 estimate becomes 0.8 points. No UP districts meet the criterion for 2015→2021. The restriction probes sensitivity to observed rotation; nonrejection does not establish random assignment.

Table C4: Separating Census Sample Selection from Covariate Adjustment

Elections	Full sample	Census sample	Census sample, adjusted
Rajasthan			
	1.4 [-0.4, 3.3]	1.8 [-0.8, 4.3]	1.7 [-0.9, 4.3]
2005–2010	$N = 4,061$	$N = 2,684$	$N = 2,684$
	-2.8 [-5.1, -0.6]	-3.1 [-5.8, -0.5]	-3.1 [-5.7, -0.4]
2010–2015	$N = 3,867$	$N = 2,343$	$N = 2,343$
	0.4 [-1.8, 2.7]	-0.2 [-3.1, 2.7]	-0.3 [-3.2, 2.6]
2015–2020	$N = 4,014$	$N = 2,279$	$N = 2,279$
Uttar Pradesh			
	2.1 [1.2, 2.9]	1.3 [0.0, 2.7]	1.2 [-0.1, 2.6]
2005–2010	$N = 28,002$	$N = 11,607$	$N = 11,607$
	1.6 [0.7, 2.6]	1.5 [0.0, 3.0]	1.7 [0.2, 3.2]
2010–2015	$N = 25,963$	$N = 9,904$	$N = 9,904$
	3.6 [2.6, 4.5]	3.7 [1.9, 5.5]	3.6 [1.7, 5.4]
2015–2021	$N = 30,975$	$N = 6,517$	$N = 6,517$

Percentage-point estimates with 95% block-clustered intervals in brackets and model sample sizes below. All columns include outcome-year block fixed effects. Adjusted models add the five 2001 Census covariates described in Section 4.2.

Table C5: Sensitivity to Block Definitions and Caste Reservation

Elections	N	Outcome block	Treatment block	Block × caste
Rajasthan				
2005–2010	4,061	1.4 [-0.4, 3.3]	1.4 [-0.4, 3.3]	1.7 [-0.2, 3.5]
2010–2015	3,867	-2.8 [-5.1, -0.6]	-2.8 [-5.1, -0.6]	-3.8 [-6.2, -1.4]
2015–2020	4,014	0.4 [-1.8, 2.7]	0.4 [-1.8, 2.7]	0.5 [-1.9, 2.9]
Uttar Pradesh				
2005–2010	28,002	2.1 [1.2, 2.9]	2.1 [1.2, 2.9]	1.8 [0.9, 2.7]
2010–2015	25,963	1.6 [0.7, 2.6]	1.6 [0.7, 2.6]	1.7 [0.8, 2.6]
2015–2021	30,975	3.6 [2.6, 4.5]	3.6 [2.6, 4.5]	3.5 [2.6, 4.5]

Percentage-point estimates with 95% block-clustered intervals in brackets. Columns use the same observations within each row. The last column absorbs treatment-year block-by-caste-reservation groups and clusters by treatment-year block.

C.4 Replication with independent UP data

Independently collected UP data from Weaver (2025), assembled for Narasimhan and Weaver (2024), give estimates of 1.5 points for 2010→2015 and 3.7 points for 2015→2021, using 37,484 and 39,222 GPs, respectively (Table C6). These estimates are similar to our main UP results. The models use 2011 Census district–block identifiers for fixed effects and clustering; Appendix A.5 describes the source and its election-year coding.

Table C6: Short-Term Effects: Weaver UP Replication

	2015 [i]	2015 (FE) [ii]	2021 [iii]	2021 (FE) [iv]
Constant	15.5***		18.1***	
	(0.3)		(0.3)	
Quota_{t-1}	1.7***	1.5***	3.8***	3.7***
	(0.4)	(0.4)	(0.4)	(0.4)
R^2	0.00	0.05	0.00	0.05
Observations	37,484	37,484	39,222	39,222
Block fixed effects	No	Yes	No	Yes

***p<0.01; **p<0.05; *p<0.1. The dependent variable is whether a woman was elected in an open seat in the subsequent election. Columns [i]-[ii] show effects of 2010 quota on 2015 outcomes; columns [iii]-[iv] show effects of 2015 quota on 2021 outcomes (coded 2020 in the source). Models [i] and [iii] use 2011 Census district–block clustered SE; the fixed-effects models use the same clustering. Models [ii] and [iv] add 2011 Census district–block FE. Coefficients and standard errors are in percentage points; intercepts are percentages.

D Additional Reserved Terms

Adding a second reserved term yields small, imprecise estimated increments. Larger full-sample estimates for three terms rather than one are sensitive to the districts included. We distinguish these additional-term comparisons from three terms versus no observed reservation, which also includes the immediately preceding reserved term.

D.1 History contrasts, sample support, and precision

Table D1 combines the contrasts, history counts, shared blocks, inference, and approximate minimum detectable effects. The three digits encode reservation in 2005, 2010, and 2015: 1 indicates a women's quota and 0 its absence. Thus 001 denotes reservation only in 2015 and 000 denotes no reservation in the three observed prior elections. Outcomes are Rajasthan's 2020 and UP's 2021 elections. Contrasts use the full covariance of the reservation-history model. Bootstrap intervals use 9,999 null-imposed wild cluster draws; the minimum detectable effects use a normal approximation at 80% power and two-sided 5% size.

Table D1: Reservation-History Contrasts: Support, Inference, and Precision

State	Sample	History	N (L/R)	Mixed blocks	Estimate	Cluster CI	Bootstrap CI	MDE
Rajasthan	full	111-000	149/225	65	9.4	[2.9, 15.9]	[2.8, 16.0]	9.2
Rajasthan	full	111-001	149/873	88	5.3	[-1.0, 11.7]	[-1.0, 11.7]	9.0
Rajasthan	full	101-001	411/873	166	3.4	[-0.6, 7.4]	[-0.7, 7.4]	5.7
Rajasthan	full	011-001	273/873	128	1.9	[-3.1, 6.8]	[-3.2, 6.8]	7.1
Rajasthan	restricted	111-000	58/77	31	6.7	[-2.9, 16.3]	[-3.1, 16.5]	13.4
Rajasthan	restricted	111-001	58/153	37	-2.4	[-12.3, 7.5]	[-12.3, 7.6]	13.7
Rajasthan	restricted	101-001	62/153	33	-2.0	[-13.2, 9.2]	[-13.5, 9.2]	15.6
Rajasthan	restricted	011-001	94/153	37	-1.2	[-10.4, 8.1]	[-10.6, 8.0]	12.8
UP	full	111-000	798/2916	385	7.7	[4.5, 10.8]	[4.5, 10.9]	4.6
UP	full	111-001	798/3978	392	4.0	[0.8, 7.2]	[0.8, 7.1]	4.5
UP	full	101-001	2398/3978	533	-0.3	[-2.4, 1.8]	[-2.4, 1.8]	3.0
UP	full	011-001	1279/3978	489	1.2	[-1.5, 3.8]	[-1.5, 3.8]	3.7
UP	restricted	111-000	149/426	65	8.4	[1.2, 15.6]	[1.2, 15.6]	10.1
UP	restricted	111-001	149/602	68	1.4	[-5.8, 8.6]	[-5.9, 8.6]	10.1
UP	restricted	101-001	349/602	84	-0.9	[-6.5, 4.7]	[-6.5, 4.6]	7.9
UP	restricted	011-001	203/602	75	-1.5	[-7.2, 4.2]	[-7.1, 4.3]	8.0

Note:
Estimates, 95% intervals, and approximate minimum detectable effects (MDEs) are in percentage points. All models include block fixed effects and cluster by block. Mixed blocks contain both compared groups; L/R counts refer to the left/right groups in each contrast. Model samples: Rajasthan full 2,671; Rajasthan restricted 601; UP full 19,264; UP restricted 2,952 . Bootstrap intervals use 9,999 null-imposed wild cluster draws. MDEs use a normal approximation at 80% power and two-sided 5% size.

D.2 Sensitivity to district restrictions

The restricted rows retain districts where tests do not detect dependence between quota assignments in either 2005–2010 or 2010–2015. The samples fall to 601 GPs in Rajasthan and 2,952 in UP. No UP districts satisfy the stricter restriction covering all three transitions. Nonrejection does not establish random assignment; the restriction probes sensitivity to the observed rotation pattern.

The separate comparison of three reserved terms with no observed reservation yields 9.4 percentage points in Rajasthan (95% interval [2.8, 16.0]) and 7.7 in UP ([4.5, 10.9]). Under the district restriction, the estimates are 6.7 points in Rajasthan ([-3.1, 16.5]) and 8.4 in UP ([1.2, 15.6]). These contrasts include the immediately preceding reservation, so they do not isolate the gain from extending reservation beyond one term.

D.3 Replication with independent UP data

The independent UP panel described in Appendix A.5 supports comparisons of 2010 and 2015 reservation histories at the 2021 election (Table D2). Its shorter coverage does not permit the three-prior-term comparison.

Table D2: Long-Term Effects: Weaver UP Replication

	No FE [i]	FE [ii]
Constant	17.9***	
	(0.4)	
Quota_{2010}	0.4	0.2
	(0.5)	(0.5)
Quota_{2015}	3.6***	3.6***
	(0.5)	(0.5)
$\text{Quota}_{2010} \times \text{Quota}_{2015}$	0.7	0.6
	(1.0)	(1.0)
R^2	0.00	0.05
Observations	36,925	36,925
Block fixed effects	No	Yes

***p<0.01; **p<0.05; *p<0.1. The dependent variable is whether a woman was elected in an open seat in 2021 (coded 2020 in the source). All models include 2010 quota, 2015 quota, and their interaction. Model [i] uses 2011 Census district–block clustered SE; the fixed-effects model uses the same clustering. Model [ii] adds 2011 Census district–block FE. Coefficients and standard errors are in percentage points; intercepts are percentages.

E Persistence across Subsequent Open Elections

E.1 Data and comparisons

The duration estimand in Section 4.1 compares women's electoral success across consecutive open elections following a fixed reservation history, without renewed reservation. It uses the four-election panels described in Appendix A: 2005, 2010, 2015, and 2020 in Rajasthan, and 2005, 2010, 2015, and 2021 in Uttar Pradesh. The outcome is whether a woman wins the GP leadership election. This exploratory analysis estimates duration separately from next-election effects and cumulative exposure.

We first compare GPs reserved for women in 2005 and open in all subsequent elections with GPs open in all four elections. We denote these histories 1000 and 0000, where 1 indicates a women's quota and 0 its absence. Caste reservations may remain in either group. This comparison follows 329 GPs in Rajasthan and 6,825 in UP through three consecutive open elections. Two further comparisons follow GPs reserved in 2010, either without reservation in 2005 (0100) or with it (1100), through the two subsequent open elections. Each uses 0000 as its comparison history. Reservation before 2005 is unobserved, so 0000 does not mean never exposed to a quota.

Within each comparison, we require observed winner sex at every follow-up. The same GPs therefore contribute to all points on a trajectory. The samples differ across reservation histories, and the comparison groups overlap. We estimate a separate linear probability model at each follow-up, with 2005 district–samiti fixed effects in Rajasthan and district–block fixed effects in UP. Standard errors cluster at these baseline units. Only blocks containing both histories identify the within-block contrast; Table E1 reports their number alongside sample sizes and unadjusted winner shares.

For a history comparison a, let S_a denote its fixed analysis sample, $n_{d,a}$ the number of GPs in block d, and $p_{d,a}$ the exposed share in that block. The coefficient in Equation 2 estimates

$$\theta_{a,k} = \sum_d w_{d,a} \left\{ E[Y_{t(a,k)} \mid \text{Exposed}_a = 1, d, S_a] - E[Y_{t(a,k)} \mid \text{Exposed}_a = 0, d, S_a] \right\}, \tag{3}$$

with $w_{d,a}$ proportional to $n_{d,a} p_{d,a}(1 - p_{d,a})$ and normalized to sum to one over blocks containing both histories. These weights remain fixed across follow-ups because the GP sample and exposure indicator do not change. Thus $\theta_{a,k} - \theta_{a,1}$ compares the same weighted histories at different elections. It is not a comparison of different treatment intensities, and the fixed-effects coefficient is not generally an equally weighted average across GPs.

Random initial assignment alone need not make every subsequent history comparison exchangeable. Conditioning on later open status can preserve randomization when future assignment depends only on past assignments and the controlled strata. A causal interpretation additionally requires that future eligibility, panel linkage, and missing outcomes do not select the compared histories on other predictors of women's electoral success. We therefore report conditional history contrasts. Within a trajectory, elapsed time also coincides with the calendar election; the analysis does not identify a general rate of decay separate from election-specific conditions.

E.2 Results and changes over time

Figure E1 displays the estimates at each open election. For the 2005 reservation comparison, UP's estimates remain small at all three open elections: 1.5, -0.2, and 1.2 percentage points. Rajasthan's estimates are -1.2, -7.7, and 0.6 points. The negative Rajasthan estimate at the second open election is followed by an estimate close to zero at the third. Its wide intervals permit both positive and negative differences at the first and third elections.

Rajasthan's other histories show larger changes. For 0100 versus 0000, the estimate rises from -7.2 points at the first open election to 9.6 at the second (95% interval [4.5, 14.9]). For 1100 versus 0000, it rises from -12.6 to 9.1 points ([1.9, 16.4]). The corresponding UP estimates are 0.9 and -0.2 points for 0100, and 2.3 and 1.0 for 1100. Thus the data do not support a uniform account in which positive effects appear immediately and then fade.

We test changes directly by regressing each GP's later-minus-first outcome on the history indicator with the same fixed effects. This preserves the covariance between repeated observations. Pointwise intervals and two-sided tests use 9,999 null-imposed Rademacher wild cluster bootstrap draws. Table E2 reports both unadjusted and Holm-adjusted p-values for the eight temporal comparisons across the two states and three histories. The two Rajasthan increases following 2010 reservation remain distinguishable from zero after adjustment. The third-minus-first differences following 2005 reservation are 1.8 points in Rajasthan (95% interval [-8.5, 12.1]; $p = 0.733$) and -0.3 in UP ([-2.8, 2.2]; $p = 0.828$). These intervals do not establish that persistence is constant. Comparing the significance of separate election estimates would not test whether they differ.

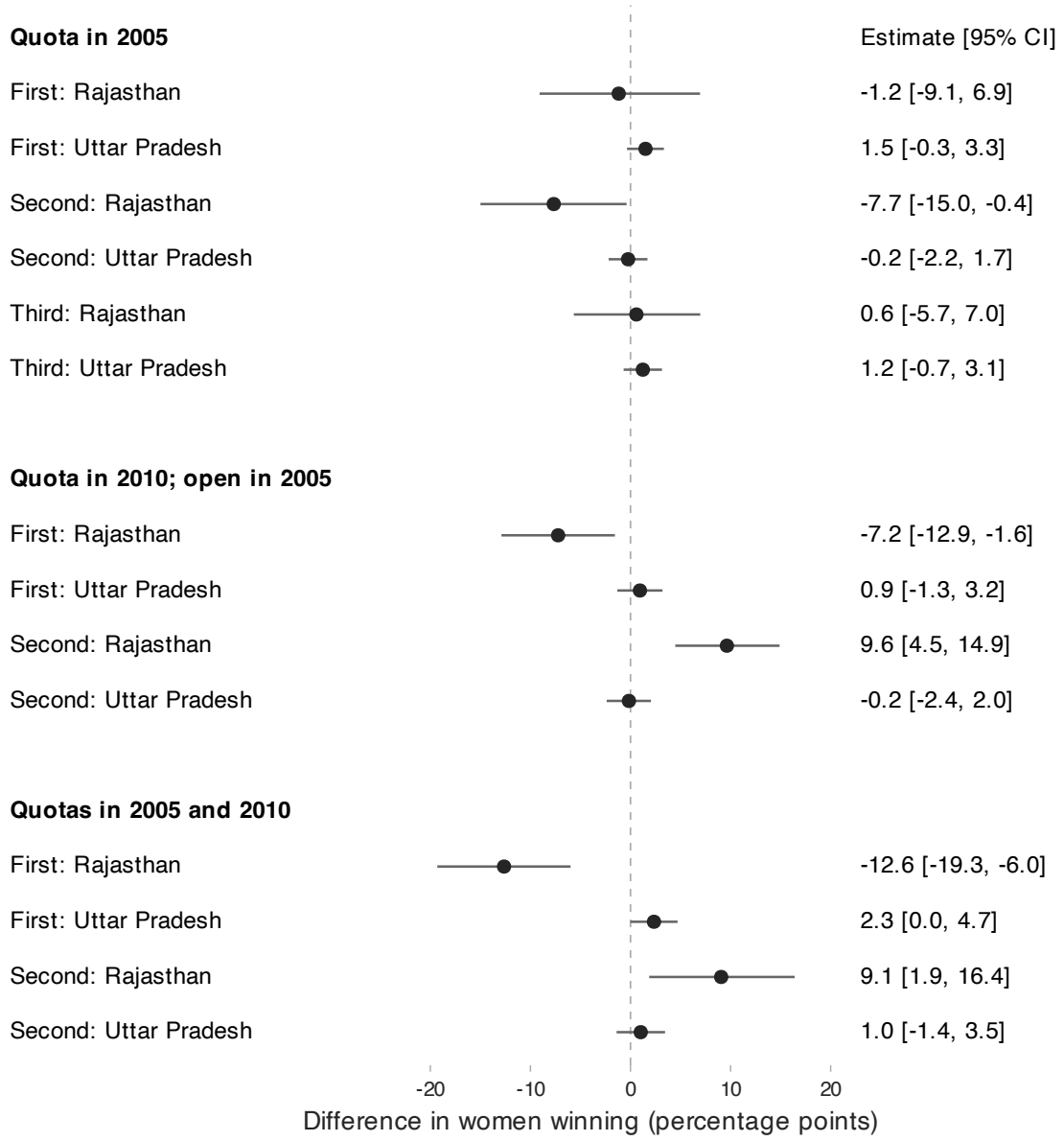


Figure E1: Women's electoral success across consecutive open elections

Notes: Each group compares the indicated reservation history with offices open in all four observed elections. No women's reservation intervenes during follow-up. Within each state and reservation history, the same GPs contribute at every follow-up; samples differ across histories. Points are block fixed-effects contrasts in the probability that a woman wins, in percentage points. Horizontal lines are pointwise 95% wild cluster bootstrap intervals. For the first history, follow-up elections are 2010, 2015, and 2020 (Rajasthan) or 2021 (UP); for the other histories, they are 2015 and 2020 or 2021. Sources: the linked Rajasthan and UP election panels described in Appendix A.

Table E1: Reservation histories and later open-election outcomes

State	History	Election	N (T/C)	Mixed blocks	Women T/C (%)	Estimate (pp)	95% interval
Rajasthan	1000–0000	2010	104/225	58	8.7/11.1	-1.2	[-9.1, 6.9]
Rajasthan	1000–0000	2015	104/225	58	6.7/12.0	-7.7	[-15.0, -0.4]
Rajasthan	1000–0000	2020	104/225	58	6.7/8.4	0.6	[-5.7, 7.0]
Rajasthan	0100–0000	2015	417/225	96	6.5/12.0	-7.2	[-12.9, -1.6]
Rajasthan	0100–0000	2020	417/225	96	13.2/8.4	9.6	[4.5, 14.9]
Rajasthan	1100–0000	2015	219/225	75	4.1/12.0	-12.6	[-19.3, -6.0]
Rajasthan	1100–0000	2020	219/225	75	15.1/8.4	9.1	[1.9, 16.4]
Uttar Pradesh	1000–0000	2010	3911/2914	539	15.3/14.2	1.5	[-0.3, 3.3]
Uttar Pradesh	1000–0000	2015	3911/2914	539	15.7/15.4	-0.2	[-2.2, 1.7]
Uttar Pradesh	1000–0000	2021	3911/2914	539	17.9/17.1	1.2	[-0.7, 3.1]
Uttar Pradesh	0100–0000	2015	1986/2916	507	16.8/15.4	0.9	[-1.3, 3.2]
Uttar Pradesh	0100–0000	2021	1986/2916	507	16.8/17.0	-0.2	[-2.4, 2.0]
Uttar Pradesh	1100–0000	2015	1997/2916	509	18.0/15.4	2.3	[0.0, 4.7]
Uttar Pradesh	1100–0000	2021	1997/2916	509	18.5/17.0	1.0	[-1.4, 3.5]

Notes: Histories list women's reservation in chronological order across the four elections: 1 denotes reserved, 0 open. T denotes the indicated exposure history; C denotes 0000. Sample sizes and unadjusted female-winner shares refer to the same complete-case GPs at every follow-up within a history comparison. Mixed blocks contain both histories. Estimates include baseline block fixed effects; intervals are pointwise 95% wild cluster bootstrap intervals. All estimates and intervals are in percentage points. Sources: linked state election panels.

Table E2: Changes in history contrasts across open elections

State	History	Comparison	Change (pp)	95% interval	Bootstrap p	Holm p
Rajasthan	1000–0000	Second minus first	-6.5	[-17.9, 4.8]	0.270	1.000
Rajasthan	1000–0000	Third minus first	1.8	[-8.5, 12.1]	0.733	1.000
Rajasthan	0100–0000	Second minus first	16.9	[9.8, 23.9]	< 0.001	< 0.001
Rajasthan	1100–0000	Second minus first	21.7	[11.9, 31.6]	< 0.001	< 0.001
Uttar Pradesh	1000–0000	Second minus first	-1.7	[-4.3, 0.8]	0.183	1.000
Uttar Pradesh	1000–0000	Third minus first	-0.3	[-2.8, 2.2]	0.828	1.000
Uttar Pradesh	0100–0000	Second minus first	-1.1	[-4.1, 1.9]	0.469	1.000
Uttar Pradesh	1100–0000	Second minus first	-1.3	[-4.4, 1.8]	0.405	1.000

Notes: Each row tests the change in a history contrast using paired outcomes from the same GPs. A positive value means a larger difference at the later election. Models include baseline block fixed effects and cluster at that level. Intervals are pointwise 95% wild cluster bootstrap intervals; Holm adjustment covers all eight temporal comparisons in this table. Adjustment concerns changes over time, not all level estimates in Figure E1.

E.3 Sensitivity

Table E3 separates two sample choices for the 2005 comparison. First, allowing each election's available outcomes within the same histories gives similar estimates to the complete-case analysis. Second, requiring open status only through the election being studied retains more GPs at earlier follow-ups. These estimates also remain small in UP. In Rajasthan, the first-election estimate becomes slightly positive and the second-election estimate becomes less negative. These changing samples cannot by themselves establish how an

effect changes over time. The table uses cluster-t intervals for all specifications to keep their uncertainty estimates comparable.

The larger changes in Rajasthan's 0100 and 1100 comparisons do not depend on any single block: removing each block in turn leaves the first-election estimates negative and the second-election estimates positive. This influence check does not resolve the selection assumptions for the histories. The trajectories also share comparison GPs and answer a different question from the next-election contrasts, so they do not enter Appendix G's next-election meta-analysis as additional independent studies.

Table E3: Sensitivity to follow-up and outcome requirements

State	Sample	Election	N (T/C)	Estimate (pp)	Cluster 95% interval
Rajasthan	Same GPs	2010	104/225	-1.2	[-9.1, 6.7]
Rajasthan	Available outcomes	2010	105/229	-2.1	[-10.1, 5.9]
Rajasthan	Same GPs	2015	104/225	-7.7	[-14.9, -0.4]
Rajasthan	Available outcomes	2015	105/229	-7.5	[-14.6, -0.4]
Rajasthan	Same GPs	2020	104/225	0.6	[-5.5, 6.7]
Rajasthan	Available outcomes	2020	104/225	0.6	[-5.5, 6.7]
Rajasthan	Open through followup	2010	924/1916	1.4	[-0.9, 3.8]
Rajasthan	Open through followup	2015	287/619	-3.6	[-7.3, 0.0]
Rajasthan	Open through followup	2020	104/225	0.6	[-5.5, 6.7]
Uttar Pradesh	Same GPs	2010	3911/2914	1.5	[-0.3, 3.3]
Uttar Pradesh	Available outcomes	2010	3911/2914	1.5	[-0.3, 3.3]
Uttar Pradesh	Same GPs	2015	3911/2914	-0.2	[-2.2, 1.7]
Uttar Pradesh	Available outcomes	2015	3912/2916	-0.2	[-2.2, 1.7]
Uttar Pradesh	Same GPs	2021	3911/2914	1.2	[-0.7, 3.1]
Uttar Pradesh	Available outcomes	2021	3912/2916	1.3	[-0.7, 3.2]
Uttar Pradesh	Open through followup	2010	9311/10251	1.3	[0.3, 2.4]
Uttar Pradesh	Open through followup	2015	6537/5645	0.0	[-1.4, 1.4]
Uttar Pradesh	Open through followup	2021	3912/2916	1.3	[-0.7, 3.2]

Notes: All rows compare reservation versus no reservation in 2005. "Same GPs" requires open status and observed outcomes at all three subsequent elections. "Available outcomes" retains the same reservation histories but requires an outcome only at the displayed election. "Open through followup" requires no reservation between 2005 and the displayed election, allowing the sample to vary over time. N (T/C) reports exposed/comparison counts. Estimates include baseline block fixed effects and block-clustered t intervals, in percentage points. Sources: linked state election panels.

F Women's Candidacy and Electoral Support

Table F1 reports candidacy and electoral-support outcomes for Rajasthan GPs open to women and men in 2020. The models include quota assignments in 2005, 2010, and 2015 and all their interactions; even-numbered columns add block fixed effects. The main-text comparison uses these columns to contrast reservation only in 2015 with no reservation in the three observed prior elections. The coefficient on the 2015 assignment identifies that contrast because the two earlier assignments are zero in both histories. Its estimates provide no clear evidence of increased candidacy or electoral support, but the intervals permit meaningful gains. Comparisons involving multiple reserved terms require sums of the relevant coefficients, rather than interpreting individual interaction coefficients as effects of those histories. The vote outcome includes only votes received by women among the winner and runner-up, divided by all valid votes.

Table F1: Candidacy in Open Seats (2020, Rajasthan)

	Women (%)		Women (N)		Any woman (%)		Votes (%)	
	No FE	FE	No FE	FE	No FE	FE	No FE	FE
	[i]	[ii]	[iii]	[iv]	[v]	[vi]	[vii]	[viii]
Constant	11.8***		0.8***		44.5***		7.3***	
	(1.6)		(0.1)		(4.0)		(1.4)	
$Quota_{2005}$	-0.1	1.6	0.0	0.1	-6.4	-4.4	-1.0	0.5
	(2.5)	(2.2)	(0.2)	(0.1)	(5.9)	(5.5)	(2.0)	(1.8)
$Quota_{2010}$	-0.3	0.9	-0.1	0.0	-0.7	1.2	0.6	2.1
	(1.5)	(1.4)	(0.1)	(0.1)	(4.5)	(4.2)	(1.4)	(1.3)
$Quota_{2015}$	-0.8	0.7	-0.1	0.0	-2.8	-0.7	0.6	2.1
	(1.6)	(1.3)	(0.1)	(0.1)	(4.1)	(3.9)	(1.4)	(1.3)
$Quota_{2005} \times Quota_{2010}$	1.9	0.3	0.1	0.0	7.9	5.6	3.1	1.9
	(3.0)	(2.9)	(0.2)	(0.2)	(6.9)	(6.7)	(2.6)	(2.6)
$Quota_{2005} \times Quota_{2015}$	1.3	-0.6	0.1	0.0	10.2	6.6	3.5	1.7
	(2.8)	(2.5)	(0.2)	(0.2)	(6.4)	(6.1)	(2.3)	(2.2)
$Quota_{2010} \times Quota_{2015}$	1.0	-0.3	0.1	0.0	2.2	-0.1	0.4	-1.2
	(2.1)	(1.9)	(0.1)	(0.1)	(5.7)	(5.5)	(1.9)	(1.8)
$Quota_{2005} \times Quota_{2010} \times Quota_{2015}$	-1.6	0.4	-0.1	0.1	-10.7	-7.0	-2.5	-0.6
	(3.6)	(3.3)	(0.2)	(0.2)	(8.5)	(8.3)	(3.2)	(3.0)
R^2	0.00	0.22	0.00	0.21	0.00	0.17	0.01	0.21
Observations	2,723	2,723	2,723	2,723	2,723	2,723	2,619	2,619
Block fixed effects	No	Yes	No	Yes	No	Yes	No	Yes

The sample comprises GPs open in 2020 with observed treatment histories. Candidate records are linked within the same general-election phase and GP. The vote outcome is votes for women among the top two candidates divided by all valid votes; it does not include votes for women placed third or lower. Unresolved candidate counts or sex are unavailable where needed for the outcome. Standard errors clustered by district–samiti. Even columns add district–samiti fixed effects. Rate outcomes use percentage points for coefficients and standard errors; intercepts are percentages. Candidate-count outcomes use candidates per GP.

G Evidence across Settings and Overall Meta-analysis

The pooled next-election association across 7 settings is 1.5 percentage points (95% interval [-1.5, 4.5]). This appendix documents the sources, component estimates, pooling, and sensitivity. Comparisons involving longer reservation histories or survey officeholding are reported separately. Appendix H tests rural–urban differences. Differences in assignment, reservation histories, and office type limit the causal interpretation of the synthesis.

G.1 Data sources and samples

Each comparison concerns an office in which women can hold office without a women's quota in the outcome election. The outcome is an indicator for a woman winning or being selected for that office. The exposure is prior reservation for women; the comparison group consists of offices without the corresponding prior exposure. Caste reservations may remain in either group. Conditioning on the later absence of a women's quota can select offices according to prior treatment when reservation rotates. We therefore distinguish published estimates based on an assignment design from descriptive comparisons of reservation histories.

Rajasthan and Uttar Pradesh. The main electoral panels and their construction are described in Section A. We retain the full-sample fixed-effects estimates for the three adjacent-election comparisons in each state for the next-election synthesis. Appendix D reports the cumulative-history contrasts for the final election. The adjacent-election estimates compare reservation in the preceding election without fixing earlier histories, so they do not isolate exactly one lifetime term of exposure. Repeated elections share offices and do not constitute independent studies.

Mumbai. We reproduce the 1997–2002 comparison in Bhavnani (2009), using the original replication data. The sample consists of 118 wards with largely unchanged boundaries that were not reserved for women in 2002. The exposure is women's reservation in 1997. Wards without a treatment classification in the published comparison are excluded. Table G1 reports the female-winner counts and denominators.

Delhi. We link the 2007 and 2012 municipal elections using the candidate data assembled by Goyal (2025). The Delhi State Election Commission's January 2012 reservation order retained the wards established in February 2007 while assigning them to three municipal corporations. Ward identifiers therefore provide a one-to-one link for all 272 wards. We verify the earlier reservation categories against the 2007 official order and reconcile abbreviated ward names against that roster. The outcome sample contains 134 offices not reserved for women in 2012.

We use the research dataset's winner-sex field, checked against the Delhi government's municipal directory and the Association for Democratic Reforms' women-winners list. The review corrects two

conflicting codes; one additional outcome has conflicting source descriptions and remains missing.[3] No outcome is assigned from the ward's reservation status or from the winner's name alone.

Delhi's 2012 order assigns caste reservations by population ranking and women's reservations by alternating positions in ordered ward lists. Conditioning on a ward being open in 2012 therefore does not produce an unrestricted randomized comparison of its 2007 reservation. The Delhi estimates are descriptive.

Chandigarh. We link the 2006 and 2011 municipal elections using the State Election Commission's official election report. The administration's October 7, 2011 letter retains the earlier 26-ward geography (PDF p. 15). The later non-quota sample contains 17 wards: nine previously reserved for women and eight not previously reserved. Winner sex comes from sex-specific honorifics in the separate 2011 candidate list, linked within ward to the election results. The 2006 report includes a later by-election replacement; we retain the original general-election winner and the deferred ward-7 election belonging to that council cohort.

The October 14, 2011 directions assign women's reservations by public lottery, excluding wards previously reserved for women until rotation is complete (PDF pp. 19–20). Caste reservations follow rankings of the 2001 Scheduled Caste population share; women-reserved SC wards have a separate draw. These rules establish lottery assignment but also select the later non-quota sample. We report the conditional contrast and adjustment for prior and current caste reservation.

The 2016 election report documents the same ward boundaries (PDF pp. 9–10), a fresh women's reservation draw, and updated Scheduled Caste population rankings. Seventeen wards were not reserved for women: six reserved in 2011 and eleven not reserved in 2011. We again link winner names to independently printed candidate honorifics. Four of the eleven comparison wards had been reserved in 2006; a sensitivity analysis excludes them. The two election comparisons contribute one city average to the synthesis.

Bengaluru. Original State Election Commission winner gazettes and the 2015 reservation notification establish 198 wards under the July 2009 boundary schedule. Of 101 wards not reserved for women in 2015, 25 were reserved in 2010 and 76 were not. We review the reservation categories against the original documents and retain the final reservation notification where it conflicts with the later winner gazette. This review corrects secondary-source women's reservation flags in two wards that affect the comparison frame.

Winner sex uses the independently recorded Gender field in the OpenCity councilor roster, checked against all winner-gazette titles. A non-sex-specific title is corroborated against the same person's title in the official ward-budget roster. Conflicting records remain unresolved, leaving 1 missing outcome in the later non-quota sample. The statutory rotation requirement and stable boundaries do not establish

[3]The Directorate of Information and Publicity's 2016 directory identifies the same ward and officeholder in each corrected record: Raj Kumari in ward 111 is recorded as a woman, and Mahak Singh in ward 254 as a man. The unresolved record is ward 115: the research file codes Dimple Chadha as female, whereas contemporaneous reporting uses male pronouns. We retain both possible values in the missing-outcome sensitivity.

a lottery for the observed assignments. We therefore interpret the Bengaluru estimates as reservation-history comparisons and examine adjustment for prior and current caste reservation.

Urban Uttar Pradesh. We link the Uttar Pradesh State Election Commission's 2012 and 2017 municipal council and town council head-election records. Links require the same district and office type and exact body names after standardizing Unicode and spacing; ambiguous identities are excluded. The sample contains 395 offices not reserved for women in 2017, with known 2012 reservation and separately recorded winner sex. These are council heads, whereas the Mumbai, Delhi, Chandigarh, and Bengaluru samples concern ward representatives. Matching names and verified source records do not establish unchanged boundaries or random prior reservation within the later non-quota sample. We report this descriptive comparison separately from the pooled estimates.

Bihar. Bihar provides a comparison of women's election as Mukhiya in 2016 according to the office's reservation in 2011. The 2009 amendment to the Bihar Panchayat Raj Act, section 4, extended reservation to two consecutive elections beginning in 2006. Thus the 2011 exposure falls in the second election of that reservation cycle. The comparison captures withdrawal under a two-election regime; it is not a replication of the one-term municipal exposure.

We link historical State Election Commission Mukhiya winner reports to the 2016 candidate results and reservation categories within district and block. The linkage yields 3,342 offices. Repeated identifiers are excluded, and uncertain prior categories remain missing. Among offices with known reservation in both elections, 1,310 were not reserved for women in 2016; 1,295 have an observed winner-sex outcome across 220 blocks. We recover outcomes from supplemental official winner lists only when the office is uniquely identified or the listed winner agrees with a reviewed, unambiguous vote ranking. The outcome uses explicit candidate-sex fields. Mukhiya, the village-head office, is distinct from Bihar's Sarpanch, who heads the village court.

Rule 14 of the Bihar Panchayat Election Rules assigns women's reservations by population ordering within reservation categories, rather than by an unrestricted lottery. Consequently, the estimates describe a selected set of reservation transitions and do not establish a causal effect of two reserved terms. The 2016–2021 pair does not provide an additional withdrawal comparison because reservation continued through that two-election cycle.

West Bengal appointments. Beaman et al. (2009, Table III) report female Pradhan appointments in Birbhum, Burdwan, Hooghly, Howrah, Nadia, and South 24 Parganas. Elected council members select the Pradhan. Their sample excludes offices reserved for women in the outcome year. The 2003 regression compares offices reserved in 1998 with other offices. The 2008 regression jointly distinguishes reservation in 1998 only, 2003 only, and both years, with neither year reserved as the reference category. Regressions include district fixed effects and heteroskedasticity-robust standard errors. The six districts were selected for logistical proximity, so the results do not represent a random statewide sample. We use the published coefficients and standard errors; these are not estimates recomputed from the underlying microdata.

Birbhum follow-up survey. We also analyze the reservation roster and Pradhan survey associated with Beaman et al. (2009), checking the earlier reservation history against the Chattopadhyay and Duflo (2004) data. The roster contains 165 GPs; 151 have verified histories, of which 99 were not reserved for women in 2003. Sex is observed for 94 current Pradhans in the 2006 survey. Conflicting questionnaire fields remain missing. This measures officeholding during the term, rather than a verified 2003 appointment, and can differ if officeholders were replaced. Because Birbhum is also included in the published six-district analysis, the survey comparison is supplementary evidence on measurement and timing, not an independent replication.

Table G1: Samples for the additional microdata comparisons

Setting	Prior quota	Eligible	Observed	Women	Women (%)
Bengaluru 2010–2015	Not reserved	76	76	3	3.9
Bengaluru 2010–2015	Reserved	25	24	0	0.0
Chandigarh 2006–2011	Not reserved	8	8	0	0.0
Chandigarh 2006–2011	Reserved	9	9	1	11.1
Chandigarh 2011–2016	Not reserved	11	11	0	0.0
Chandigarh 2011–2016	Reserved	6	6	0	0.0
Delhi 2007–2012	Not reserved	90	89	2	2.2
Delhi 2007–2012	Reserved	44	44	3	6.8
Mumbai 1997–2002	Not reserved	81	81	3	3.7
Mumbai 1997–2002	Reserved	37	37	8	21.6
UP urban 2012–2017	Not reserved	199	199	33	16.6
UP urban 2012–2017	Reserved	196	196	42	21.4
Bihar 2011–2016	Not reserved	534	527	72	13.7
Bihar 2011–2016	Reserved	776	768	128	16.7
Birbhum 1998–2006	Not reserved	69	66	11	16.7
Birbhum 1998–2006	Reserved	30	28	5	17.9

Notes: Eligible offices have known prior reservation and are not reserved for women in the outcome election. Observed offices additionally have a resolved officeholder-sex outcome. Women's shares use observed offices as the denominator. Mumbai uses the original study's classified, largely stable-boundary wards. Chandigarh and Delhi link ward identifiers under explicit boundary-retention orders; urban UP links council heads by district, office type and body name; Bihar links 2011 and 2016 Mukhiya records. Birbhum links 1998 reservation to 2006 survey officeholding among offices open in 2003. Sources and linkage rules are described above. Published West Bengal appointment estimates lack corresponding arm-level microdata in this analysis; their regression sample sizes appear in the figures.

G.2 Next-election comparisons

Figure G1 reports the next-election estimates and their pooled summary. Rajasthan and UP retain the main analysis's full-sample fixed-effects estimates and wild cluster bootstrap intervals. For Mumbai, Delhi, Chandigarh, and Bengaluru, we calculate raw differences in proportions with Newcombe intervals combining the two Wilson score intervals (Newcombe, 1998). These intervals remain nondegenerate when neither group has a woman winner. The Mumbai counts and point estimate reproduce the published com-

parison, while this interval differs from the original article's one-sided proportion tests. For published West Bengal estimates, we reconstruct normal intervals from the rounded robust standard errors.

Mumbai's difference is 17.9 percentage points (95% interval [5.7, 33.7]), compared with 4.6 points in Delhi ([-2.6, 16.1]). These are differences in female-winner probabilities among offices open to women without a women's quota, not differences in compliance with a current quota.

In Chandigarh's 2011 election, one woman won among the nine previously reserved wards, compared with none among eight previously nonreserved wards. The difference is 11.1 percentage points (95% interval [-22.6, 43.5]). In 2016, no woman won in either group: the difference is 0.0 points ([-25.9, 39.0]). Both comparisons are imprecise.

In Bengaluru, women won 0 of 24 previously reserved wards with resolved outcomes and 3 of 76 previously nonreserved wards. The difference is -3.9 percentage points (95% interval [-11.0, 10.1]). These additional urban comparisons provide no evidence of a common large positive gain.

Urban UP also yields an imprecise positive difference. Women won 21.4% of later non-quota offices previously reserved for women, compared with 16.6% of previously non-quota offices. The difference is 4.8 percentage points (95% interval [-3.8, 13.6]), giving each office equal weight and using a wild cluster bootstrap across 18 administrative divisions. Adjustment for district, office type and prior caste reservation gives 5.7 points (division-clustered 95% interval [-5.7, 17.1]). Both intervals permit zero and appreciable positive differences.

Women account for 9.2% of Pradhan appointments in the West Bengal comparison group in 2003 and 10.9% in 2008. Those offices were unreserved in the observed prior elections. The one-term difference in 2003 is 2.7 percentage points (95% interval [-1.8, 7.2]). In 2008, the recent one-term contrast, which holds 1998 reservation absent, is 0.3 points ([-4.8, 5.4]).

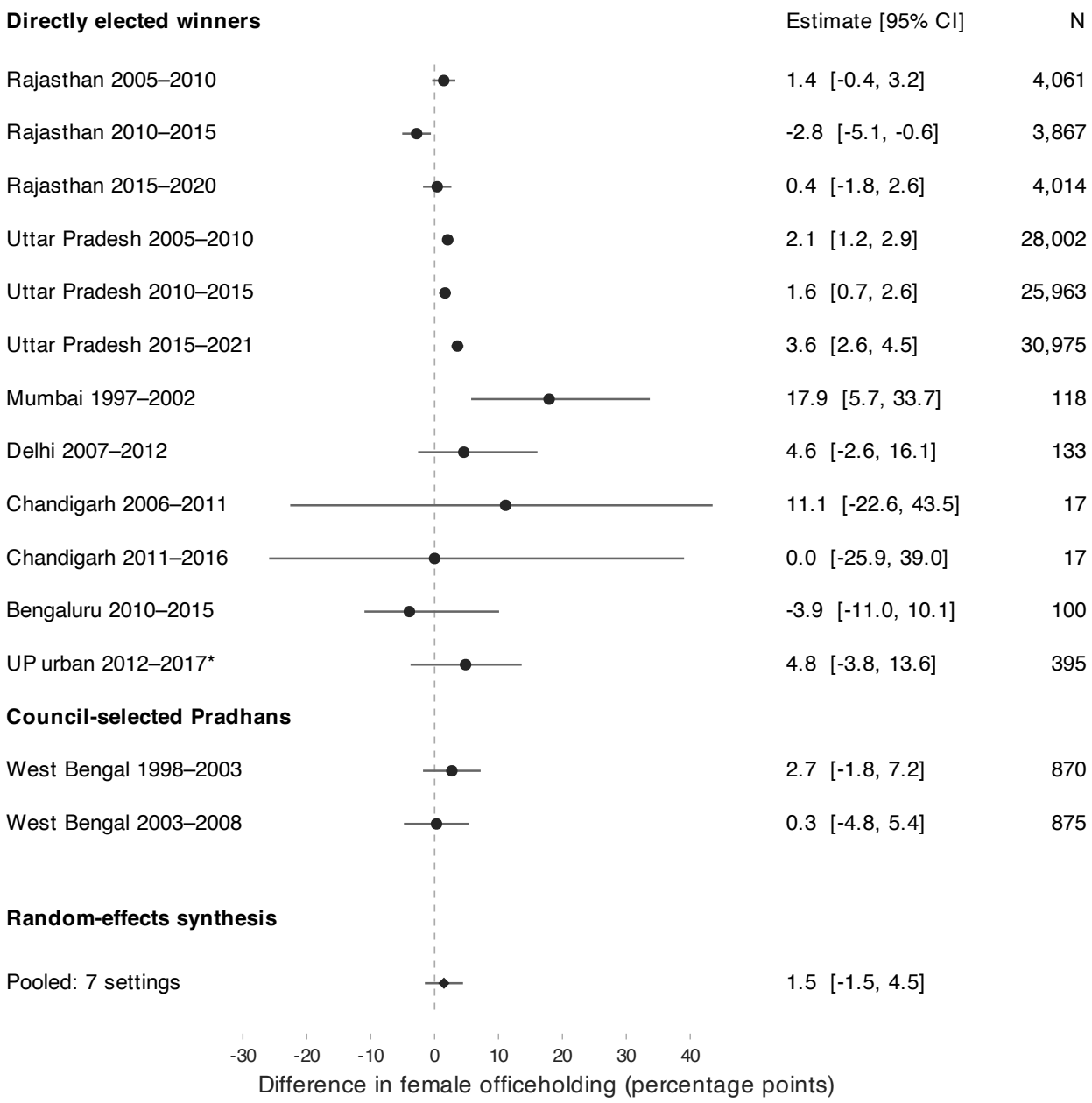


Figure G1: Next-election comparisons and pooled persistence estimate

Notes: Points compare prior women-reserved and other offices among offices not reserved for women in the outcome election. Lines show 95% intervals. Rural Rajasthan and UP use full-sample fixed-effects estimates and wild cluster bootstrap intervals; Mumbai, Delhi, Chandigarh, and Bengaluru use raw differences and Newcombe–Wilson intervals. Urban UP uses an unadjusted difference and a division-level wild cluster bootstrap with 9,999 draws. Its asterisk marks exclusion from the pooled estimate. West Bengal uses published district-adjusted estimates and normal intervals reconstructed from rounded robust standard errors (Beaman et al., 2009, Table III, columns 1–2). Its 2008 row compares 2003-only reservation with neither 1998 nor 2003 reservation. The diamond summarizes seven setting averages using Section G.3. *N* is each model's office count; overlapping samples must not be summed.

G.3 Overall meta-analysis

We combine the next-election comparisons in rural Rajasthan, Uttar Pradesh, and West Bengal with Mumbai, Delhi, Chandigarh, and Bengaluru. Within Rajasthan and Uttar Pradesh, we give each of the three election comparisons equal weight; within West Bengal and Chandigarh, we average two comparisons. Mumbai, Delhi, and Bengaluru each contribute one estimate. This produces one average per geographic setting, so repeated elections do not count as independent studies. Urban UP remains outside the synthesis for the reasons described in Section G.1. Bihar enters a sensitivity analysis because its reservation lasts for two elections. The Birbhum survey is excluded from pooling because its timing differs and it overlaps the published West Bengal sample.

The urban binary outcomes require care in estimating sampling variances. An ordinary plug-in variance is zero when both groups have no woman winner, as in Chandigarh in 2016. For every urban ward comparison, we retain the raw difference but regularize its variance: with x women among n offices in each arm, we use $\tilde{p} = (x + 1/2)/(n + 1)$ and sum $\tilde{p}(1 - \tilde{p})/n$ across the two arms. A sensitivity analysis instead adds one to each cell. These approximate variances determine synthesis weights; they are distinct from the score intervals displayed for individual comparisons and are not guaranteed to be conservative.

The published West Bengal estimates do not report their covariance. For a common dependence sensitivity analysis across settings, we vary the assumed correlation between election estimates; a separate specification uses the estimated cross-election covariances for Rajasthan and UP from Appendix C.2. For a setting with m estimates and standard errors s_j, we calculate the variance of its arithmetic mean as

$$v(\rho) = \frac{(1-\rho)\sum_{j=1}^{m} s_j^2 + \rho\left(\sum_{j=1}^{m} s_j\right)^2}{m^2}.$$

The primary calculation sets $\rho = 1$, the upper bound on the sampling variance of this mean given the marginal standard errors. These are assumptions about dependence, not estimates of it. Maximizing within-setting sampling variance does not guarantee conservative inference for the subsequent random-effects model.

We estimate the between-setting variance by restricted maximum likelihood and weight setting averages by the inverse of their sampling variance plus this estimated variance. The 95% interval uses a modified Knapp–Hartung adjustment that cannot reduce the unadjusted standard error and a t critical value with the number of settings minus one degrees of freedom (Viechtbauer, 2010). Sampling errors are assumed independent across settings. With few settings, the pooled interval and heterogeneity estimate remain sensitive to model assumptions. This exploratory synthesis covers the assembled evidence rather than a systematic sample of studies or Indian local governments.

The pooled difference is 1.5 percentage points (95% interval [-1.5, 4.5]), with an estimated between-setting standard deviation of 1.5 points. The average point estimate is positive, but the interval includes zero and permits both a negative association and a larger positive association.

G.4 Sensitivity of the pooled estimate

Table G2 varies the within-setting correlation from the primary value of one to zero, an intermediate value, and the lowest feasible equicorrelation, $-1/(m-1)$. It also adds Bihar's two-election regime, omits each setting in turn, and retains only the earliest or latest comparison per setting. Omitting West Bengal restricts the synthesis to directly elected offices. These choices alter dependence assumptions or the population of comparisons; they do not supply additional independent replications.

Omitting one setting at a time produces pooled point estimates in the range [0.4, 2.4] points. The primary interval includes zero, but the interval excluding Rajasthan and the interval using only the earliest comparison per setting exclude zero. The average association is therefore estimated positively, while its precision depends on geographic and temporal composition. Pooling does not resolve the selection and measurement concerns of the component comparisons.

Table G2: Sensitivity of the pooled next-election comparison

Specification	Settings	Difference	95% interval
Primary: maximum within-setting correlation	7	1.5	[-1.5, 4.5]
Within-setting correlation = 0	7	1.4	[-1.5, 4.2]
Within-setting correlation = 0.5	7	1.4	[-1.5, 4.4]
Minimum feasible within-setting correlation	7	2.2	[-1.2, 5.7]
Estimated covariance in Rajasthan and UP	7	1.4	[-1.7, 4.4]
Urban variance correction = 1	7	1.6	[-1.3, 4.4]
Include Bihar's two-election regime	8	1.7	[-0.7, 4.0]
Omit Rajasthan	6	2.4	[0.9, 3.8]
Omit Uttar Pradesh	6	0.4	[-3.2, 4.1]
Omit Mumbai	6	1.2	[-1.4, 3.7]
Omit Delhi	6	1.3	[-2.2, 4.8]
Omit Chandigarh	6	1.4	[-2.0, 4.9]
Omit Bengaluru	6	1.8	[-1.3, 4.9]
Omit West Bengal	6	1.5	[-2.8, 5.8]
Earliest comparison per setting	7	2.0	[0.8, 3.1]
Latest comparison per setting	7	2.0	[-1.6, 5.5]

Notes: Differences and 95% intervals are in percentage points. All rows use REML and modified Knapp–Hartung inference. Except where varied, election estimates are averaged equally within setting with correlation one. The estimated-covariance row uses the joint block-cluster covariance for Rajasthan and UP and retains correlation one for West Bengal and Chandigarh. The minimum feasible correlation is $-1/2$ for three comparisons and -1 for two; single-comparison settings are unchanged. Omitting West Bengal retains only directly elected offices. Including Bihar adds an eighth setting with a two-election reservation regime. Earliest and latest rows retain one comparison per setting. These are overlapping sensitivity analyses.

G.5 Other reservation histories and follow-up measures

Figure G2 presents comparisons excluded from the primary next-election synthesis because they concern different histories or outcome timing. Bihar's two-election reservation regime yields an unadjusted difference of 3.0 percentage points (95% interval [-1.1, 7.1]), using block-clustered HC1 standard errors and a t critical value based on the number of blocks minus one.

In West Bengal, exposure in both 1998 and 2003 yields an estimated 7.6 percentage points at the 2008 election (95% interval [-0.4, 15.6]). That larger point estimate is imprecise and does not establish greater persistence than one reserved term. The figure also reports 1998-only exposure at the 2008 election; both contrasts use neither year reserved as the reference.

In the Birbhum survey, women hold office in 11 of 66 comparison GPs (16.7%) and 5 of 28 previously reserved GPs (17.9%). The unadjusted difference associated with 1998 reservation is 1.2 percentage points (95% interval [-16.1, 18.5]). This interval is too wide to establish that persistence is small. The Birbhum survey interval uses HC2 inference. Its sample overlaps the published West Bengal analysis, so it is not an independent study.

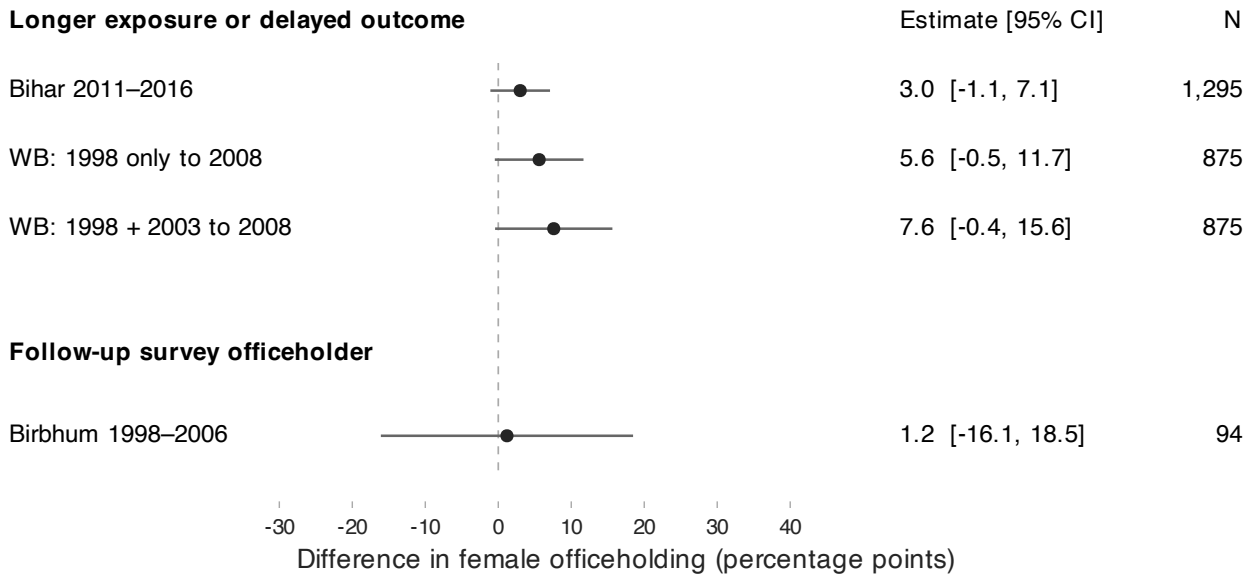


Figure G2: Other Reservation Histories and Follow-up Officeholding

Notes: The scale matches Figure G1. Bihar's 2011 exposure belongs to a two-election reservation regime; its unadjusted difference uses block-clustered inference. West Bengal (WB) rows compare the indicated history with neither year reserved, using published estimates and reconstructed normal intervals. Birbhum measures the current Pradhan in a 2006 survey, with an HC2 interval. All intervals are 95%. The West Bengal comparisons share samples, and Birbhum overlaps the published West Bengal study. None of these rows enters the primary pooled estimate.

G.6 Sensitivity to adjustment, linkage, and missing outcomes

In Delhi, assigning the unresolved outcome each possible binary value bounds the difference at [3.5, 4.6] points over the complete eligible roster. The corresponding Bengaluru bounds are [-3.9, 0.1] points. These are finite-roster bounds, not confidence intervals. Table G3 reports caste adjustments and restrictions to unchanged caste categories. We also exclude the two Bengaluru wards with conflicting primary and secondary quota codes. Such restrictions change the population being compared; restricting on later caste categories does not establish exogeneity.

In Bihar, adjustment for block fixed effects and prior caste reservation reduces the difference to 1.4 points (95% interval [-3.3, 6.2]). Restricting the linkage to exact agreement of normalized, transliterated village names produces a smaller, imprecise estimate. Name agreement establishes a narrower linkage criterion; it does not independently establish unchanged village boundaries. The 15 missing outcomes bound the difference within the eligible roster at [1.7, 4.0] points. These bounds do not cover the 530 linked open offices excluded for unknown prior reservation, the incomplete geographic coverage of the historical reports, or linkage errors.

In Birbhum, assigning each of the 5 missing outcomes both possible values bounds the difference within the verified roster at [-3.6, 7.4] points. Adjustment for prior caste and block, and then for current caste, also yields imprecise differences. The latter adjustment conditions on a later reservation category. These finite-roster missing-outcome bounds are not confidence intervals and do not incorporate sampling uncertainty or selection into an office being open.

Table G3: Sensitivity to adjustment and sample definition

Setting	Specification	Difference	95% interval	*N*
Bengaluru 2010–2015	Unadjusted	-3.9	[-11.0, 10.1]	100
Bengaluru 2010–2015	Prior and current caste	-3.3	[-7.5, 0.9]	100
Bengaluru 2010–2015	Unchanged caste category	0.0	[-16.8, 56.1]	22
Bengaluru 2010–2015	Exclude conflicting quota codes	-4.1	[-11.3, 10.0]	98
Bihar	Unadjusted	3.0	[-1.1, 7.1]	1,295
Bihar	Block and prior caste	1.4	[-3.3, 6.2]	1,295
Bihar	Name score = 1	1.7	[-3.7, 7.0]	815
Bihar	Candidate results only	3.4	[-0.7, 7.4]	1,289
Birbhum	Unadjusted	1.2	[-16.1, 18.5]	94
Birbhum	Prior caste and block	2.5	[-18.4, 23.3]	94
Birbhum	Also current caste	0.7	[-20.6, 22.0]	94
Chandigarh 2006–2011	Unadjusted	11.1	[-22.6, 43.5]	17
Chandigarh 2006–2011	Prior and current caste	13.4	[-15.8, 42.7]	17
Chandigarh 2011–2016	Unadjusted	0.0	[-25.9, 39.0]	17
Chandigarh 2011–2016	Exclude 2006 quota exposure	0.0	[-35.4, 39.0]	13
Delhi 2007–2012	Unadjusted	4.6	[-2.6, 16.1]	133
Delhi 2007–2012	Prior SC and corporation	4.5	[-3.8, 12.7]	133
Delhi 2007–2012	Unchanged SC category	5.1	[-3.9, 14.1]	123

Notes: Differences and 95% intervals are in percentage points. Unadjusted urban differences and the urban sample restrictions use Newcombe–Wilson intervals. Adjusted urban models and Birbhum use HC2 standard errors and residual-degrees-of-freedom *t* intervals. Bihar clusters by district–block with the HC1 finite-sample adjustment and *t* critical values based on the number of blocks minus one. Missing prior caste is a separate category in Bihar's adjusted model. "Name score = 1" requires exact agreement of normalized, transliterated village names; "Candidate results only" excludes recovered outcomes. The Chandigarh 2011 and Bengaluru adjustments include both prior and current caste reservation. The Chandigarh 2016 outcome is zero in every eligible ward, so adjustment cannot identify any further outcome variation. The unchanged-caste Delhi restriction and the Birbhum adjustment for current caste condition on later reservation categories. Specifications within a setting overlap and do not count as independent studies.

H Rural–Urban Differences and Development

The rural point estimate is lower than the urban estimate, but the difference is imprecise and sensitive to which cities are included. Figure G1 also reports the descriptive urban UP comparison; the meta-analysis here retains Mumbai, Delhi, Chandigarh, and Bengaluru as its urban settings. Comparisons within rural Rajasthan and UP also do not establish a consistent relationship between development and subsequent electoral gains.

H.1 Rural–urban meta-analysis

We test whether the prior-reservation contrast is smaller in rural settings using a meta-regression of setting-average persistence estimates on an intercept and a rural indicator. Mumbai, Delhi, Chandigarh, and Bengaluru are urban; Rajasthan, Uttar Pradesh, and West Bengal are rural. The model jointly estimates the subgroup means and their difference, using the same within-setting averages and assumed sampling variances as Section G.3. It estimates a common residual between-setting variance by REML and uses modified Knapp–Hartung inference with the number of settings minus two degrees of freedom. The coefficient tests equality of the persistence contrasts, rather than equality of the underlying female-officeholding rates or separate significance within each subgroup.

The fitted rural and urban persistence contrasts are 1.3 and 2.6 percentage points, respectively. Their difference, rural minus urban, is -1.3 points, with a 95% interval of [-11.3, 8.7] and a two-sided p-value of 0.745. The rural point estimate is lower, but the interval permits both a substantial rural disadvantage and an advantage. The test establishes neither a difference nor equivalence.

H.2 Sensitivity of the rural–urban difference

Table H1 applies the dependence and sample-composition changes defined above to the rural–urban coefficient. Adding Bihar supplies a fourth rural setting under a longer reservation regime. Omitting West Bengal restricts both groups to directly elected offices. Omitting each city in turn tests its influence, leaving three urban settings. The urban variance correction adds one rather than one-half to each binary-outcome cell when estimating sampling variances, retaining the raw differences. The earliest- and latest-election specifications assess temporal composition without counting repeated elections as independent settings.

The confidence interval includes zero in every reported specification. Completing both unresolved urban outcomes in all four possible ways gives rural–urban point estimates from -4.2 to -0.8 points; each interval also includes zero. The evidence is compatible with lower rural persistence, but the small number of settings leaves the contrast imprecise. Geography also coincides with differences in the office being contested, reservation rules, and election periods. Neither the primary subgroup analysis nor these sensitivity checks identify an effect of urbanization.

Table H1: Direct tests of the rural–urban persistence difference

Specification	Rural − urban	95% interval	p
Primary: maximum within-setting correlation	-1.3	[-11.3, 8.7]	0.745
Within-setting correlation = 0	-1.5	[-11.3, 8.3]	0.711
Within-setting correlation = 0.5	-1.4	[-11.3, 8.5]	0.732
Minimum feasible within-setting correlation	-3.3	[-9.9, 3.3]	0.255
Estimated covariance in Rajasthan and UP	-1.5	[-11.5, 8.6]	0.721
Urban variance correction = 1	-2.2	[-12.4, 8.0]	0.598
Include Bihar's two-election regime	-0.9	[-9.6, 7.8]	0.812
Omit Rajasthan	-1.6	[-16.3, 13.0]	0.773
Omit Uttar Pradesh	-3.3	[-18.4, 11.8]	0.578
Omit Mumbai	1.3	[-7.4, 10.1]	0.695
Omit Delhi	-0.3	[-14.6, 14.0]	0.957
Omit Chandigarh	-1.2	[-13.5, 11.1]	0.798
Omit Bengaluru	-6.8	[-17.7, 4.0]	0.156
Omit West Bengal	-2.7	[-17.5, 12.2]	0.641
Earliest comparison per setting	-0.4	[-9.3, 8.5]	0.910
Latest comparison per setting	-0.8	[-11.4, 9.7]	0.848

Notes: Differences and 95% intervals are in percentage points; negative estimates indicate lower persistence in rural settings. The primary model has three rural and four urban settings. Each row fits an intercept and rural indicator with a common residual heterogeneity variance, REML estimation, and modified Knapp–Hartung $t(k-2)$ inference. Reported p-values are two-sided tests of a zero rural–urban difference. The correlation and sample changes match Table G2; adding Bihar supplies a fourth rural setting. Omitting each city is an influence diagnostic with three urban settings remaining. These exploratory comparisons do not identify a causal effect of urbanization.

H.3 Development within rural settings

We interact prior quota assignment with three 2001 Census characteristics measured through SHRUG: proximity to towns, infrastructure availability, and female literacy. All three interactions enter the same model with outcome-year block fixed effects and clustering.

Near Town indicates a GP at or below the median distance to the nearest town. High Infrastructure indicates an above-median sum of education, medical, electricity, and banking availability; presence requires at least one linked village with the facility, while absence requires observed absence in every constituent village. High Female Literacy indicates an above-median share of women aged seven or older who are literate. Cutoffs use matched GPs with observed covariates before restriction to the outcome sample; missing measures remain missing.

Table H2 reports the interaction estimates using mean village-to-town distance, followed by sensitivity to using the minimum. Neither definition establishes a consistent relationship between development and the quota effect. The intervals permit meaningful differences.

Table H2: Development and Quota Effects: Primary Estimates and Distance Sensitivity

State	Elections	Characteristic	Mean distance	Minimum distance
Rajasthan	2005–2010	Near town	-2.3 [-7.2, 2.6]	0.0 [-4.7, 4.6]
Rajasthan	2005–2010	High infrastructure	-0.7 [-6.5, 5.0]	-0.9 [-6.6, 4.9]
Rajasthan	2005–2010	High female literacy	3.5 [-1.3, 8.3]	3.2 [-1.6, 8.0]
Rajasthan	2010–2015	Near town	-0.5 [-5.4, 4.4]	-0.1 [-4.9, 4.7]
Rajasthan	2010–2015	High infrastructure	1.6 [-3.2, 6.4]	1.7 [-3.1, 6.5]
Rajasthan	2010–2015	High female literacy	0.5 [-4.4, 5.4]	0.5 [-4.4, 5.4]
Rajasthan	2015–2020	Near town	4.2 [-1.5, 9.9]	1.6 [-3.9, 7.1]
Rajasthan	2015–2020	High infrastructure	-4.8 [-11.3, 1.8]	-4.7 [-11.2, 1.8]
Rajasthan	2015–2020	High female literacy	5.3 [-0.4, 11.0]	5.8 [0.0, 11.5]
Uttar Pradesh	2005–2010	Near town	1.1 [-1.4, 3.7]	1.2 [-1.4, 3.8]
Uttar Pradesh	2005–2010	High infrastructure	2.3 [-0.4, 4.9]	2.3 [-0.4, 4.9]
Uttar Pradesh	2005–2010	High female literacy	0.3 [-2.2, 2.8]	0.3 [-2.2, 2.8]
Uttar Pradesh	2010–2015	Near town	1.0 [-2.2, 4.2]	1.3 [-1.9, 4.6]
Uttar Pradesh	2010–2015	High infrastructure	0.1 [-3.0, 3.2]	0.1 [-3.0, 3.2]
Uttar Pradesh	2010–2015	High female literacy	-1.3 [-4.5, 2.0]	-1.3 [-4.6, 2.0]
Uttar Pradesh	2015–2021	Near town	-1.0 [-4.8, 2.7]	-1.8 [-5.7, 2.2]
Uttar Pradesh	2015–2021	High infrastructure	-1.7 [-5.8, 2.3]	-1.7 [-5.8, 2.3]
Uttar Pradesh	2015–2021	High female literacy	-1.1 [-4.8, 2.6]	-1.1 [-4.7, 2.6]

Notes: Interaction coefficients and 95% block-clustered intervals are in percentage points. Each coefficient is the difference in the quota effect between the indicated group and its comparison group. All models use outcome-year block fixed effects and include the three moderators and their interactions with prior quota assignment. The columns change only the GP distance measure used to define proximity to towns. Medians use observed covariates before restriction to the outcome sample.

I Phone Audit

The rural audit records officeholder contact in 9.9% of quota seats with an answered call and 76.7% of non-quota seats. This appendix documents the protocol, detailed responses, classification uncertainty, and the urban extension.

I.1 Sampling and protocol

The completed rural audit covers 500 representatives from women-reserved seats and 507 representatives from non-quota seats. The sampling procedure stratified both groups by district and reservation category. The retained initial selections contain 578 quota-seat and 593 non-quota-seat records; the records do not document the reduction to the completed audit samples. Mobile phones are used for citizen engagement in local politics (Rigillo, 2018); interviews with political staff also informed the audit.[4] We called the self-reported numbers in the Rajasthan State Election Commission's 2020 winner records, making up to three attempts between 10:00 am and 6:30 pm in 2024. Rural quota-seat calls lasted about one minute on average.

We used the following script for our survey, conducted by a native Hindi speaker in India. The interviewer documented the date, time, and duration of each call. To monitor data quality, interviewers provided screenshots of call history, which we randomly sampled and matched against spreadsheet entries with perfect accuracy.

> Hello, I am [RA name] talking on behalf of researchers based in the United States who want to study the panchayat system in Rajasthan. Am I talking to [representative name]?
>
> *Notes to RA: If the name does not match, ask them to transfer the call to the actual representative. If they refuse, thank them and end the conversation, recording their reason.*
>
> If the actual representative answers: We obtained your phone number from the Rajasthan Election Commission's website. We want to understand governance in panchayats in Rajasthan. We have a few questions about your experience as an elected representative; this will take 2-3 minutes. We will not share any personal information and are not recording this call, but will take notes. Would you be willing to talk to us?

If the respondent consents, proceed to questions:

1. Have you contested in elections before?
2. Has anybody from your family contested in elections before?
3. Are the Panchayat meetings held regularly?
4. Do you attend meetings regularly?

[4] Author's phone interview with former secretarial staff (name and affiliation withheld on request), December 20, 2024.

I.2 Rural audit results

Table I1: Detailed Phone Audit Responses

Panel A: Gender-Quota Seats

Category	N (%)
Contact attempts	
Answered	382 (76.4)
No answer recorded	118 (23.6)
Among answered	
Officeholder contact recorded	38 (9.9)
Male non-member only	321 (84.0)
Female non-member only	6 (1.6)
Non-member, sex unavailable	6 (1.6)
Identity unavailable	11 (2.9)
Recorded male non-members	
Child	46 (14.3)
Other recorded relative	45 (14.0)
Relationship recorded as self	1 (0.3)
Relationship unavailable	56 (17.4)
Spouse	174 (54.0)
Recorded non-members	
Did not transfer	286 (85.6)
Transfer recorded as not applicable	2 (0.6)
Transfer status unavailable	43 (12.9)
Transferred	3 (0.9)

Panel B: Non-Quota Seats

Category	N (%)
Contact attempts	
Answered	404 (79.7)
No answer recorded	103 (20.3)
Among answered	
Officeholder contact recorded	310 (76.7)
Male non-member only	65 (16.1)
Female non-member only	10 (2.5)
Non-member, sex unavailable	3 (0.7)
Identity unavailable	16 (4.0)

Notes: Rajasthan village heads elected in 2020, contacted in 2024. The unit is a sampled officeholder. Contact includes transfers and callbacks; answered wrong numbers count as answered contacts. Officeholder contact takes precedence over other categories. Relationship and transfer panels include all recorded non-member respondents, including those who subsequently passed the call to the officeholder. Percentages use the denominator indicated by each panel.

I.3 Sensitivity to respondent classification

Table I2: Officeholder Contact: Sensitivity to Respondent Classification

Sample	Answered	Recorded contact (%)	Classification range (%)
Rural quota	382	9.9	[9.4, 13.4]
Rural open	404	76.7	[74.3, 80.9]
Jaipur quota	66	33.3	[28.8, 34.8]
Jaipur open	143	90.9	[90.9, 93.7]

Notes: Percentages use sampled officeholders with at least one answered call as the denominator. Recorded contact includes transfers and callbacks. The classification range treats unidentified respondents and unresolved identity conflicts as unknown, then assigns every unknown contact to a non-member (lower endpoint) or the officeholder (upper endpoint). Conflicts include disagreement between a claimed member's recorded sex and the uniquely linked elected officeholder's sex, and contradictory respondent fields or notes. Neither sex nor identity is silently adjudicated. These ranges describe classification uncertainty among answered records, not sampling uncertainty or outcomes for unanswered records.

I.4 Characteristics of reached officeholders

The 38 quota-seat officeholders recorded as reached by phone are a selected subset of quota winners. Their mean age is 36.9 years, compared with 41.3 among quota winners overall, and a smaller share report being unemployed (Table I3). Recorded contact includes transfers and callbacks; it does not establish who exercises political authority.

Table I3: Characteristics of Representatives with Recorded Phone Contact

Characteristic	Mean	Observed N
Age	36.9	38
Graduate (%)	36.1	36
Unemployed (%)	55.3	38
Log assets	12.5	38

Note:
Quota-seat officeholders recorded as reached on at least one attempt, including transfers and callbacks. N = 38

I.5 Jaipur extension

Table I4: Urban Phone Survey Response Distribution (Jaipur)

(a) Gender-Quota Seats

Category	N (%)
Contact attempts	
Answered	66 (79.5)
No answer recorded	17 (20.5)
Among answered	
Officeholder contact recorded	22 (33.3)
Male non-member only	40 (60.6)
Female non-member only	2 (3.0)
Non-member, sex unavailable	1 (1.5)
Identity unavailable	1 (1.5)

(b) Open Seats

Category	N (%)
Contact attempts	
Answered	143 (86.7)
No answer recorded	22 (13.3)
Among answered	
Officeholder contact recorded	130 (90.9)
Male non-member only	7 (4.9)
Female non-member only	0 (0.0)
Non-member, sex unavailable	2 (1.4)
Identity unavailable	4 (2.8)

Notes: Jaipur Greater and Jaipur Heritage, 2025. Calls were attempted to 83 quota-seat and 165 open-seat representatives. Reservation status is taken from the official ward category (State Election Commission, Rajasthan, 2020, pp. 43–53). The unit and contact definitions match Table 3.

J Candidate and Officeholder Characteristics

Quota-seat winners are younger and have lower average education and reported assets than non-quota winners. The women-only Rajasthan comparisons show similar graduation rates and mean log assets across reservation categories. These comparisons describe the people selected under the two regimes, rather than measuring governing ability.

J.1 Rajasthan measurement and samples

The Rajasthan comparisons use the State Election Commission's 2020 candidate records linked to winners within the same GP and election phase. Candidate and event identifiers must be unique; winners additionally require a unique name match to the winner record. Age and occupation come from the candidate records. Graduation includes graduate, postgraduate, professional graduate, and professional postgraduate categories; missing education is excluded from that outcome. Reported unemployment is the recorded occupation category and need not imply an absence of work. We winsorize assets at the 10th and 90th percentiles of the candidate sample and take the natural logarithm of one plus assets.

For both states, ages outside 21–110 are treated as missing for age comparisons; other recorded characteristics remain usable. The lower bound follows the minimum age for office (Government of Rajasthan, 1994a; Government of Uttar Pradesh, 1947); the upper bound is a plausibility screen for recording errors. Table 4 compares all winners and candidates by reservation status. Table J1 restricts each group to recorded women. This supplementary comparison helps interpret composition; it does not replace the comparison with all non-quota officeholders.

Table J1: Candidate Characteristics Among Women: Quota vs. Open Seats

	Women winners			Women candidates		
Characteristic	Open	Quota	Difference	Open	Quota	Difference
Age	42.8	41.3	1.6	40.6	39.9	0.7
Graduate (%)	12.2	12.7	-0.5	17.3	13.3	4.0
Unemployed (%)	53.1	61.1	-8.0	54.8	61.6	-6.8
Log assets	12.6	12.6	0.0	12.1	12.1	0.0

Note:
Descriptive open-minus-quota comparisons, conditional on candidacy or winning. Graduation and unemployment are percentages; their differences are percentage points. Other rows use their stated units. Graduation rates exclude records with missing education. Age means exclude recorded ages outside 21–110 years; other outcomes retain those records. Assets winsorized at the 10th and 90th percentiles of the candidate sample before taking logs. These comparisons do not identify a causal effect on candidate quality.

Table J2: Winner Characteristics: Quota vs. Open Seats (Uttar Pradesh)

	2010			2015			2021		
Characteristic	Open	Quota	Difference	Open	Quota	Difference	Open	Quota	Difference
Age	41.7	40.3	1.4	41.5	40.3	1.1	42.2	40.9	1.3
Education	2.9	1.7	1.2	2.7	1.7	1.0	2.9	2.0	0.8
Assets (asinh)	–	–	–	–	–	–	13.0	12.9	0.1

Note:
Descriptive comparisons using the Weaver data. Differences are open minus quota. The last wave is coded 2020 in the source labels and corresponds to the 2021 election. Assets are available only in that wave. Age means exclude recorded ages outside 21–110 years; other outcomes retain those records.

J.2 Uttar Pradesh winner characteristics

Table J2 uses the independent UP records described in Appendix A.5. It compares winners' age and the source education measure in 2010, 2015, and 2021; assets are available only in the final wave and use the source's inverse-hyperbolic-sine transformation. Quota-seat winners have lower mean age and education in each election. These measures are reported in their source units and should not be compared directly with Rajasthan's graduation rate or log-assets measure.

K Do Winners Contest the Next Election?

One way quotas could increase women's subsequent representation is by helping the women elected under them build continuing political careers. We examine this route by linking Rajasthan's 2015 winners to candidates in 2020 and Uttar Pradesh's 2015 winners to candidates in 2021. This differs from the aggregate candidacy analysis in Appendix F: the outcome here concerns the previous winner, rather than all women who enter the next election.

K.1 Data and matching

We use the State Election Commission winner and candidate records and the GP links described in Appendix A. We restrict the follow-up election to seats reserved for neither women nor a caste group, so the new reservation category does not bar any prior winner from contesting. The comparison includes all prior non-quota winners, including women. Rajasthan's candidate rosters must contain the number of contestants reported in the official results and refer to a unique election event. UP's rosters must contain a single declared winner consistent with the linked panel; an independent contestant count is unavailable. Table K1 reports coverage.

Table K1: Coverage of the Winner-to-Candidate Comparison

Cohort	Prior seat	Linked	Fully open	Roster checks	Analysis
Rajasthan	Quota	3821	1202	1196	1195
Rajasthan	Non-quota	4061	707	705	704
Uttar Pradesh	Quota	15621	9918	9917	9803
Uttar Pradesh	Non-quota	30893	6561	6561	6461

Notes: Rajasthan follows 2015 winners to 2020; UP follows 2015 winners to 2021. "Fully open" means no gender or caste reservation at follow-up. Roster checks require nonmissing candidate names, unique candidate identifiers, and the state-specific checks described in the text. The analysis also requires usable prior-winner identifiers and excludes ambiguous exact matches or exact matches conflicting with recorded sex or feasible age. All matching rules use the same sample within each state.

There is no common person identifier across elections. Our strict rule requires a unique normalized name match within the linked GP; in UP it also requires the recorded father's or husband's name to match. We normalize case, punctuation, and whitespace while retaining Hindi vowel signs. For more generous matching, we transliterate names into Latin script and remove spaces. A candidate's name may then differ by up to two character edits and no more than 20% of the longer name; nonexact matches require both names to contain at least five characters. UP also has an intermediate rule requiring the relative's name to agree within three edits and 25%. The most generous rule allows a compatible candidate-name match without requiring agreement on the relative's name, and counts any such candidate as a possible return.

Recorded sex must be compatible. We also reject matches to candidates younger than 25 in Rajasthan or 26 in UP at follow-up, ages inconsistent with having reached 21 in 2015 after allowing for election dates within each year (Constitution of India, Article 243F). Prior-winner age is unavailable, so age cannot otherwise verify identity; missing follow-up age remains unknown. These rules address spelling and recording

differences without treating every similar name as a verified identity. Strict matching can miss genuine returns, while generous matching can admit namesakes. The resulting rates describe matching scenarios, not formal bounds on the true rate of contesting again.

K.2 Results

Under strict matching, 2.8% of Rajasthan's prior quota winners reappear in the next candidate roster, compared with 13.5% of prior non-quota winners. The corresponding UP rates are 10.0% and 16.5%. Adjusting for baseline block and caste reservation gives differences of -11.5 percentage points in Rajasthan (95% interval [-14.6, -8.3]) and -5.7 points in UP ([-6.9, -4.6]). These are comparisons between the officeholders selected under the two reservation regimes, not estimates of the effect of winning office on the same person's subsequent career.

Table K2: Prior Winners Matched to Candidates in the Next Election

	Matched / prior winners (%)		Adjusted difference (pp)	
Cohort	Quota	Non-quota	Estimate	95% interval
Rajasthan	34/1195 (2.8)	95/704 (13.5)	-11.5	[-14.6, -8.3]
Uttar Pradesh	983/9803 (10.0)	1069/6461 (16.5)	-5.7	[-6.9, -4.6]

Notes: Strict matching in seats fully open at follow-up. Counts and percentages are unadjusted. Adjusted differences are quota minus non-quota coefficients from linear probability models with baseline block fixed effects and baseline caste-reservation indicators. The samples contain 252 blocks in Rajasthan and 693 in UP. Intervals use block-clustered standard errors, the manuscript's finite-sample correction, and a cluster-t reference distribution. They describe sampling uncertainty conditional on the matching rule, not uncertainty about identities.

Generous matching raises the rates to 4.4% versus 20.5% in Rajasthan and 27.8% versus 47.1% in UP. Requiring a similar relative's name in UP gives intermediate rates of 22.3% and 40.7%. Thus the levels depend substantially on how identities are matched, but prior quota winners are less likely to appear under every rule (Figure K1). The comparison is consistent with limited continuation by the original quota winners. It does not exclude entry by other women, candidacy in another GP or office, or differences in matching accuracy between quota and non-quota winners.

Low rates of return also appear in earlier evidence: the 2008 Panchayati Raj Ministry report finds that over 89% of women representatives did not contest another election after their term (Ministry of Panchayati Raj, 2008, pp. 26, 63).

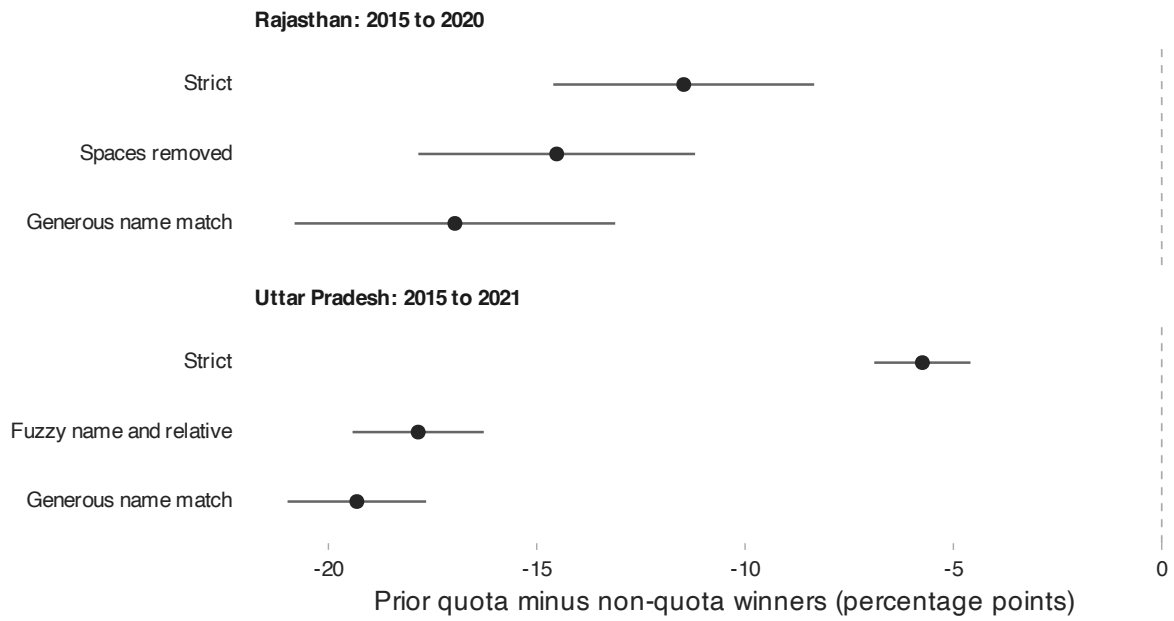


Figure K1: Sensitivity of Subsequent Candidacy Comparisons to Matching Rules

Notes: Points are adjusted differences in matched candidacy between prior quota and non-quota winners, in percentage points; lines are 95% block-clustered intervals. Each state uses a common sample across rules. Strict matching requires the name in Rajasthan and both name and relative in UP. The other rules remove spaces or allow the transliteration and edit-distance matches described in the text. The generous rule counts any compatible candidate-name match as a possible return. Zero denotes no difference. Intervals do not incorporate linkage uncertainty.